

Ethernet Gateway

User Manual

Version 0.6.3

Ricky Bryce

A telnet/SSH file-transfer gateway, SSH proxy, Hayes-compatible modem emulator (with optional telnet-serial console bridge), text-mode web browser, AI chat client, and weather service for retro hardware — XMODEM, XMODEM-1K, YMODEM, ZMODEM, Kermit, and Punter over PETSCII, ANSI, and ASCII terminals.

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1. Introduction

Ethernet Gateway is a standalone server written in Rust that bridges retro hardware and modern networks. It accepts telnet and SSH connections, speaks the XMODEM / XMODEM-1K / YMODEM / ZMODEM / Kermit / Punter file-transfer protocols, emulates a Hayes AT modem on **two physically independent serial ports** (each port also flips to a telnet-serial console bridge on demand), proxies outbound connections to remote telnet and SSH servers, renders web pages as plain text, looks up the weather, and provides a terminal AI-chat front end for the Groq API.

It was designed to let 40-year-old computers use 40-minute-old networks. The telnet interface is the headline feature because it works out of the box with any machine that can speak a serial stream: Commodore 64, CP/M systems, the AltairDuino, RC2014 / SC126, and countless other homebrew boards. Once the server is running on a PC, connect from anywhere on your LAN:

```
telnet 192.168.1.160 2323
```

Or from a serial-attached retro machine, dial through the emulator:

```
ATDT 192.168.1.160:2323
```

1.1 Supported Terminal Types

On every inbound connection the gateway asks the user to press so it can auto-detect the terminal. The byte received uniquely identifies the client:

Byte	Terminal	Typical hardware
0x14	PETSCII	Commodore 64, VIC-20, C128 in 40-column mode
0x08 or 0x7F	ANSI	Modern terminal emulators, PuTTY, xterm, macOS Terminal
anything else	ASCII	Dumb terminals, line printers, bridge boards without colour

After terminal detection the user is asked whether to enable colour. There is no default — the user must explicitly press or . Colour output uses PETSCII colour codes on the Commodore, ANSI escape sequences on modern terminals, and plain text on ASCII.

1.2 Feature Overview

- **File transfer** — XMODEM (128-byte), XMODEM-1K, YMODEM, ZMODEM (with autostart), Kermit (incl. standalone server mode), and Punter C1 (Commodore 64/128), with auto-detection on upload and a selection prompt on download.
- **Hayes AT modem emulator** — two physically independent serial ports (Port A and Port B), each with its own enabled flag, mode, baud, AT/S-register state (S0-S26), and stored phone-number slots, plus the `+++` escape.

- **Telnet-serial console bridge** — either port can be flipped to *console mode* per-port, so a telnet/SSH user can drive a microcontroller or RS-232 device directly through the **G Serial Gateway** menu (chapter 9.13). Both ports can host bridges concurrently.
- **SSH server** — optional encrypted access with its own credentials and a persistent Ed25519 host key.
- **SSH gateway** — outbound proxy for SSH servers, with TOFU host-key trust and Ed25519 public-key auth.
- **Telnet gateway** — outbound proxy with RFC-1143 Q-method option negotiation or a raw-TCP escape hatch.
- **Web browser** — text-mode HTTP/HTTPS/Gopher with forms, bookmarks, and 40-column-friendly layout.
- **AI chat** — Groq API, `llama-3.3-70b-versatile`, paginated answers wrapped to the terminal width.
- **Weather** — 3-day forecast from Open-Meteo for any US ZIP code.
- **GUI console** — eframe/egui configuration editor with live server log.

SECURITY NOTICE

Telnet transmits all data, including passwords, in cleartext. The telnet interface is intended for *local or private network use only*. Use the SSH interface for anything leaving your LAN, and keep `egateway.conf` at mode `0600` because it contains plaintext credentials.

2. Installation

2.1 Linux AppImage (pre-built)

The simplest installation on a modern Linux distribution is the pre-built AppImage that ships with every release:

```
chmod +x Ethernet_Gateway-x86_64.AppImage
./Ethernet_Gateway-x86_64.AppImage
```

AppImages bundle their own libraries and run on any mainstream glibc-based distribution without installation. On first run the AppImage creates `egateway.conf` in the current working directory.

VERIFICATION

Each release publishes a SHA-256 checksum, an optional GPG signature, and a Sigstore keyless signature bound to the publisher's GitHub identity. Verify at least the SHA-256 before running any binary from the internet: `sha256sum -c ethernet-gateway-v0.6.3-...tar.gz.sha256`.

2.2 Building from Source

The project is written in pure Rust using the 2024 edition. The minimum supported compiler is Rust **1.87**.

2.2.1 Debian 13 / Ubuntu

```
sudo apt update
sudo apt install -y build-essential pkg-config cmake curl libudev-dev
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source "$HOME/.cargo/env"
```

2.2.2 Fedora / RHEL / AlmaLinux

```
sudo dnf install -y gcc gcc-c++ make cmake pkg-config curl
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source "$HOME/.cargo/env"
```

2.2.3 Arch Linux

```
sudo pacman -S --needed base-devel cmake curl
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source "$HOME/.cargo/env"
```

2.2.4 macOS

```
xcode-select --install
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh
source "$HOME/.cargo/env"
brew install cmake
```

2.2.5 Windows

1. Run the rustup installer from `https://rustup.rs`.
2. When prompted, install the Visual Studio C++ Build Tools.
3. Install CMake from `https://cmake.org/download/` or via `winget install Kitware.CMake`.
4. Open a fresh PowerShell or `cmd.exe` so rustup is on your `PATH`.

2.2.6 Verify the toolchain

```
rustc --version    # should show 1.87 or later
cargo --version
cmake --version
```

2.2.7 Compile

```
cargo build --release
```

The resulting binary is at `target/release/ethernet-gateway`. The first full build pulls a couple of hundred crates and may take 3-8 minutes depending on your machine; subsequent incremental builds are quick.

2.2.8 Key dependencies

The gateway pulls the following well-known crates as direct dependencies: `tokio` (async runtime), `russh` (SSH server and client), `serialport` (cross-platform serial I/O), `ureq` (blocking HTTP client used by the browser and AI chat), `html2text` (HTML to plain-text rendering), `url` (URL parsing for the browser), `eframe` / `egui_extras` (GUI), `rfd` (native file-open/save dialogs), `arboard` (clipboard access), `signal-hook` (POSIX signals), and `serde_json`. Full transitive list is in `Cargo.lock`.

2.3 Running as a systemd Service (Linux)

A hardened unit file is bundled at `contrib/systemd/ethernet-gateway.service`. To install:

```
# Create a dedicated unprivileged service user
sudo useradd --system --home-dir /var/lib/ethernet-gateway \
            --shell /usr/sbin/nologin ethernet-gateway
sudo install -d -m 0750 -o ethernet-gateway -g ethernet-gateway \
            /var/lib/ethernet-gateway

# Install the binary
sudo install -m 0755 target/release/ethernet-gateway /usr/local/bin/

# Install and enable the unit
sudo install -m 0644 contrib/systemd/ethernet-gateway.service \
            /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable --now ethernet-gateway.service

# Watch the log
journalctl -u ethernet-gateway -f
```

The unit ships with defensive hardening: `NoNewPrivileges`, `PrivateTmp`, `ProtectSystem=strict`, `ProtectHome`, namespace restrictions, `SystemCallFilter=@system-service`, an empty capability bounding set, a 512 MiB memory cap, and `TasksMax=256`. The service user has no shell, no home directory, and write access only to `/var/lib/ethernet-gateway`.

If you want to listen on privileged ports (telnet on 23 or SSH on 22), uncomment the `CapabilityBoundingSet=CAP_NET_BIND_SERVICE` and matching `AmbientCapabilities` lines in the unit.

`systemctl reload ethernet-gateway` sends `SIGHUP`, which the binary handles as a graceful stop. Restart the service to pick up config changes.

3. First Run & Configuration File

On the very first launch the gateway looks for `egateway.conf` in the current working directory. If the file does not exist it is created with sensible defaults and a short log message is printed. On subsequent launches the file is read, every missing key is filled in with its default, and the file is rewritten so that future editors see the full set of keys and comments.

UNIX PERMISSIONS

`egateway.conf` holds the plaintext login password (shared by telnet, SSH, and the web UI) and the Groq API key. On Unix the file is set to mode `0600` (owner-only read/write) on every write. The same applies to `dialup.conf`, `gateway_hosts`, `ethernet_ssh_host_key`, and `ethernet_gateway_ssh_key`.

3.1 File Format

The file is a simple `key = value` text file. Blank lines and lines starting with `#` are comments and ignored. Values are trimmed of surrounding whitespace. Unknown keys are silently ignored, so you can annotate your own notes as needed. Boolean values accept `true` / `false` in any case.

```
# Ethernet Gateway Configuration
telnet_enabled = true
telnet_port = 2323
ssh_enabled = false
security_enabled = false
```

3.2 Configuration Key Overview

The table below summarises every key recognised by the parser. Full details including sanitisation, range checks, and live-edit behaviour are in [Appendix A](#). The reserved-zero S-registers and stored phone-number slots are covered in chapter 9.

Key	Type	Default	Description
telnet_enabled	bool	true	Run the telnet server. Set false for SSH-only mode.
telnet_port	u16	2323	Telnet listen port. Must be ≥ 1 .
telnet_gateway_negotiate	bool	false	Outbound gateway sends proactive WILL TTYPE / WILL NAWS / DO ECHO.
telnet_gateway_raw	bool	false	Outbound gateway bypasses all IAC parsing. Supersedes negotiate.
gateway_debug	bool	false	Byte-level trace of SSH/Telnet gateway sessions. Toggle from the GUI/web General frame or the telnet Other / Serial Configuration menus; takes effect next session. EGATEWAY_GATEWAY_DEBUG env var forces it on. See §16.10.
enable_console	bool	true	Open the GUI at startup. Set false for headless systemd operation.
security_enabled	bool	false	Require username/password login on telnet, SSH, and the web UI.
username, password	string	admin, changeme	Unified login credentials — shared by telnet, SSH, and the configuration web UI. (The legacy per-protocol ssh_username / ssh_password keys were removed in v0.6.0; configs that still have them auto-migrate into this pair on first load when the unified pair is still at the factory defaults.)
ssh_enabled	bool	false	Run the SSH server on ssh_port.
ssh_port	u16	2222	SSH listen port.
web_enabled	bool	false	Run the configuration web server on web_port. Off by default; mirrors the GUI settings page in a browser. See §5.6.
web_port	u16	8080	Web-server listen port.
ssh_gateway_auth	string	password	Outbound SSH-gateway auth mode: password prompts for the remote password each connect; key uses the gateway's auto-generated Ed25519 client key (which must be installed in the remote's authorized_keys first).
gateway_role	string	standalone	Master/Slave role: standalone master slave. See §9.14.

Key	Type	Default	Description
master_accept_relays	bool	false	Master only: accept relay connections from slaves (never implied by enabling SSH).
slave_master_host	string	<i>(empty)</i>	Slave only: the master's hostname/IP.
slave_master_port	u16	2222	Slave only: the master's SSH port.
slave_master_username	string	<i>(empty)</i>	Slave only: must match the master's unified username.
slave_master_password	string	<i>(empty)</i>	Slave only: must match the master's unified password.
relay_transport	string	ssh	Relay transport; only ssh is implemented.
transfer_dir	string	transfer	Base directory for file transfers, relative to the working dir.
max_sessions	usize	50	Concurrent telnet session cap.
idle_timeout_secs	u64	900	Session idle timeout. 0 disables.
disable_ip_safety	bool	false	When true, the gateway accepts inbound telnet/SSH from any address. Default false restricts to RFC-1918 private ranges and loopback. See §5.4.
groq_api_key	string	empty	Key for AI chat. Empty disables the feature.
browser_homepage	string	http://telnetbible.com	Page loaded when the browser opens.
weather_zip	string	empty	Last-used US ZIP. Updated automatically.
verbose	bool	false	Detailed per-block / per-subpacket protocol diagnostics (XMODEM, YMODEM, ZMODEM, Kermit, Punter).
xmodem_negotiation_timeout	u64	45	Seconds to wait for the far-end sender/receiver to start. Shared by XMODEM, XMODEM-1K, and YMODEM (same protocol code path).
xmodem_block_timeout	u64	20	Per-block timeout.
xmodem_max_retries	usize	10	NAK retry cap per block.
xmodem_negotiation_retry_interval	u64	7	Seconds between C/NAK pokes during the initial handshake. Christensen's XMODEM and Forsberg's reference implementations use ~10 s; 7 s is a compromise that starts quickly but

Key	Type	Default	Description
			avoids stockpiling extras in a slow-starting sender's input buffer.
zmodem_negotiation_timeout	u64	45	Seconds to wait for the ZRQINIT / ZRINIT handshake to complete.
zmodem_frame_timeout	u64	30	Per-frame read timeout once a transfer is live (headers / subpackets).
zmodem_max_retries	u32	10	Retry cap for ZRQINIT, ZRPOS, and ZDATA frames.
zmodem_negotiation_retry_interval	u64	5	Seconds between ZRINIT / ZRQINIT re-sends during the handshake. Analogous to xmodem_negotiation_retry_interval but for ZMODEM.
kermit_negotiation_timeout	u64	300	Send-Init handshake timeout in seconds.
kermit_packet_timeout	u64	10	Per-packet read timeout in seconds.
kermit_idle_timeout	u64	300	Server-mode idle timeout between commands. 0 disables.
kermit_max_retries	u32	5	NAK / timeout retransmits per packet.
kermit_max_packet_length	u16	4096	Advertised long-packet length (10-9024).
kermit_window_size	u8	4	Sliding-window size (1-31). 1 = stop-and-wait.
kermit_block_check_type	u8	3	1 = 6-bit checksum, 2 = 12-bit checksum, 3 = CRC-16/KERMIT.
kermit_long_packets	bool	true	Advertise long-packet capability.
kermit_sliding_windows	bool	true	Advertise sliding-window capability.
kermit_streaming	bool	true	Skip per-packet ACKs on reliable links (TCP/SSH).
kermit_attribute_packets	bool	true	Send/accept A-packets carrying size, mtime, mode, resume.
kermit_repeat_compression	bool	true	Encode runs of identical bytes with the repeat-prefix.
kermit_8bit_quote	string	auto	8th-bit quoting policy: auto (when peer asks), on, or off.
kermit_resume_partial	bool	false	Resume partial uploads using A-packet disposition='R'. Off by default — a peer that ignores disposition='R' would corrupt the result.

Key	Type	Default	Description
kermit_resume_max_age_hours	u32	168	Maximum age of a partial file we'll resume (default one week).
kermit_locking_shifts	bool	false	Use SO/SI region markers instead of per-byte 8th-bit prefix when the peer advertises CAPAS_LOCKING_SHIFT.
kermit_server_enabled	bool	false	Run the standalone Kermit-server TCP listener at startup.
kermit_server_port	u16	2424	Kermit-server listen port.
allow_atdt_kermit	bool	false	Allow the serial-modem dial target ATDT KERMIT (or kermit-server) to drop into the Kermit server. Off by default — bypasses the telnet auth gate.
punter_block_size	u16	255	Punter C1 block size in bytes (8-255). 248-byte payload + 7-byte header at the default; lower toward ~40 on noisy lines.
punter_negotiation_timeout	u64	45	Seconds to wait for the file-type handshake to complete.
punter_block_timeout	u64	20	Per-block read timeout once a transfer is live.
punter_max_retries	u32	10	Retry cap per block before the session aborts.
punter_max_bad_rounds	u32	30	Consecutive corrupt-block resend rounds tolerated before giving up. Kept higher than punter_max_retries — a real C64 peer never caps its resends, so a low value would make the gateway quit first and strand the peer.
punter_hangup_on_failure	bool	false	On give-up, drop the connection (carrier) so a C64 — which C1 cannot be told to abort — sees loss-of-carrier and exits instead of hanging. Ends the whole session; off by default.
punter_negotiation_retry_interval	u64	5	Seconds between handshake re-sends while waiting for the peer.

The serial subsystem exposes **two physically independent ports: Port A and Port B**. Each has its own copy of every per-port key listed below, prefixed with `serial_a_` or `serial_b_` respectively (e.g. `serial_a_enabled`, `serial_b_baud`). The two ports are completely independent — separate manager threads, separate AT&W persistence targets, separate

console-bridge slots — so you can run a Hayes modem on one wire and a telnet-serial bridge on the other simultaneously.

LEGACY MIGRATION

A pre-dual-port `egateway.conf` using bare `serial_*` keys auto-migrates into Port A on first read; the writer always emits the dual-port form on save. Existing single-port deployments upgrade transparently, with Port B disabled by default.

Per-port key (suffix)	Type	Default	Description
enabled	bool	false	Activate this port. Pairs with mode to choose its role.
mode	string	modem	modem = run the Hayes AT command emulator on this port (chapter 9). console = expose the port via the telnet menu's G Serial Gateway A/B picker as a telnet-serial bridge (chapter 9.13).
port	string	empty	Device path (e.g. /dev/ttyUSB0, COM3).
baud	u32	9600	Baud rate. Must be ≥ 300 .
databits	u8	8	Data bits: 5, 6, 7, or 8.
parity	string	none	none / odd / even.
stopbits	u8	1	1 or 2.
flowcontrol	string	none	none / hardware / software.
echo	bool	true	Saved ATE state. Restored by ATZ.
verbose	bool	true	Saved ATV state.
quiet	bool	false	Saved ATQ state.
s_regs	string	see ch. 9	Comma-separated decimal S0-S26 values.
x_code	u8 0-4	4	Saved ATX level.
dtr_mode	u8 0-3	0	Saved AT&D mode.
flow_mode	u8 0-4	0	Saved AT&K mode.
dcd_mode	u8 0-1	1	Saved AT&C mode.
stored_0 ... stored_3	string	empty	Stored phone-number slots for AT&Zn=s / ATDSn.
petSCII_translate	bool	false	Saved AT+PETSCII state (PETSCII translation on direct-TCP dials). Also editable from the telnet, web, and GUI config surfaces.

INPUT SANITISATION

String values have embedded newlines and carriage returns stripped before being written back to disk. That means you cannot accidentally corrupt the config by pasting a multi-line value into the GUI.

4. GUI Configuration Console

Unless you have explicitly set `enable_console = false`, a GUI configuration window opens automatically when the gateway starts. It is backed by `eframe/egui` and provides live configuration editing plus a console that streams the server log in real time.

HEADLESS MODE

Systemd deployments, Docker images, or any other headless setup should run with `enable_console = false`. Otherwise the daemon will block waiting for a display server that does not exist.

Closing the GUI window does not stop the server — it continues running in headless mode. To stop the gateway, send it `Ctrl+C`, `SIGTERM`, or `SIGHUP` (see chapter 15).

4.1 Configuration Frames

The window is organised into three rows of two framed cards each, plus buttons and a console panel at the bottom:

- **Server** — toggle telnet/SSH, set the ports, set the session cap, set the idle timeout. A *More...* button reveals the Telnet Gateway options popup.
- **Security** — Require Login toggle, plus separate username and password fields for telnet and SSH.
- **File Transfer (XMODEM)** — transfer directory, XMODEM-family timeouts (negotiation, per-block, retries). The card's subtitle reads (*More for YMODEM / ZMODEM / Punter*): a *More...* button opens a popup with the full per-protocol breakdown covering the XMODEM/XMODEM-1K/YMODEM shared keys, the independent ZMODEM tunables (handshake timeout, frame timeout, retry cap, and retry interval), and the Punter C1 tunables (block size, negotiation timeout, retry interval, block timeout, max retries).
- **AI Chat, Browser, and Weather** — Groq API key (rendered as a password field), browser homepage URL, and weather ZIP.
- **Serial Port** — three-row layout for the dual-port subsystem. The header row carries an *Enabled* checkbox each for *Serial Port A* and *Serial Port B*, plus a single *Save* button on the right that persists both ports atomically and signals each manager to reconfigure. The two rows below show one port apiece: a device-path dropdown (with a refresh  button on Port A's row that re-scans and refreshes both dropdowns), a baud field, and a per-port *More...* button that opens that port's advanced popup (mode, framing, flow, full Hayes AT state). The two popups are independent so you can compare settings side-by-side.
- **General** — Verbose XMODEM Logging, Show GUI on Startup.

At the top of the window the title shows the gateway version and the local IP address that telnet clients will use to reach it. The IP is re-detected on every paint cycle, so if your DHCP lease changes the value updates automatically.

4.2 "More..." Popups

Advanced settings that are rarely changed live in popup windows opened from their frame's *More...* button.

4.2.1 Server — More

Mirrors the primary Server controls (so you can edit ports and toggles without leaving the popup) and adds two persistent outbound-gateway decisions that used to be asked interactively at dial time:

- **Telnet Gateway → Mode** — dropdown with *Telnet* or *Raw TCP*. Telnet is the default and parses IAC in both directions; Raw TCP disables the entire IAC layer for destinations that don't speak telnet at all. Sets `telnet_gateway_raw` in `egateway.conf`.
- **Negotiate TTYPE / NAWS with remote** — checkbox, enabled only when Mode is *Telnet*. Turn on when you dial real telnet servers that want cooperative option negotiation. Sets `telnet_gateway_negotiate`.
- **SSH Gateway → Auth** — dropdown with *Key* or *Password*. Password is the default because it works out of the box with most remote accounts. When Key is selected the popup also displays the gateway's public key in a read-only text box below, ready to paste into the remote's `~/.ssh/authorized_keys`. Sets `ssh_gateway_auth`.

Both decisions take effect on the next gateway connection — no server restart is needed. The session-side Gateway Configuration menu (*Configuration* → *G* from the main telnet menu) edits the same three settings from the telnet client: **T** toggles the telnet mode (Telnet / Raw TCP), **C** toggles cooperative TTYPE / NAWS negotiation (`telnet_gateway_negotiate`), and **S** toggles the SSH auth mode (Key / Password).

POPUP STYLING

Popups render on a deep forest-green panel with lighter-green text-entry fields so they're visually distinct from the navy main window. This is a UI convention — it doesn't affect any setting.

4.2.2 Serial Port A — More / Serial Port B — More

Each port has its own popup, opened by the *More...* button on that port's row in the Serial Port frame. The two popups are independent — both can be open simultaneously so you can compare Port A's and Port B's advanced settings side-by-side. Each popup contains:

- **Mode** — dropdown with *Modem (AT Command) Mode* or *Telnet-Serial Mode*. Saving from this popup reconfigures only the changed port; the other port keeps running.
- **Bits / Par / Stop / Flow** — data bits (5-8), parity (none/odd/even), stop bits (1/2), and flow control (none/hardware/software) for this port.
- **Echo (E1) / Verbose (V1) / Quiet (Q1)** — saved `ATE`, `ATV`, `ATQ` toggles for this port.
- **Result level (X)** — 0 to 4. See ch. 9 for what each level emits.
- **DTR (&D)** — 0 to 3.
- **Flow (&K)** — 0 to 4 (modem-layer; hardware-wire flow is set by the *Flow* dropdown above).

- **DCD (&C)** — 0 or 1.
- **S-Registers** — a comma-separated list of S0...S26 values, editable as free text. The format matches `serial_a_s_regs / serial_b_s_regs` in the config file.
- **Stored Phone Numbers** — four text fields for slots `&Z0` through `&Z3`. Leave empty for unset. `ATDSn` dials these.

4.2.3 File Transfer — More

Mirrors the transfer directory row from the main card and then splits the protocol settings into two clearly-labelled sections so operators aren't surprised when editing one page changes another:

- **XMODEM / XMODEM-1K / YMODEM** — a banner explains these three protocols share the same code path, and therefore the same keys. Shown: *Negotiate (s)*, *Block (s)*, *Retries*, and *Retry interval (s)* (the last is the C/NAK poke cadence, with an inline hint noting the spec suggests ~10 s).
- **ZMODEM** — independent tunables. Shown: *Negotiate (s)*, *Frame (s)*, *Retries*, and *Retry interval (s)* (ZRINIT/ZRQINIT re-send gap, default 5).
- **Punter (C1)** — independent tunables for the Commodore single-file protocol. Shown: *Block size* (8-255, default 255), *Negotiate (s)*, *Retry interval (s)*, *Block (s)*, and *Retries*.

All values apply on the next file transfer — no server restart is required. The corresponding telnet-side pages live under *Configuration* → *F (File Transfer)* → *X / Y / Z*.

4.3 Saving and Restarting

Each popup has its own **Save** button, which writes the current configuration to `egateway.conf` *without restarting the server*. This is useful for changes that take effect on the next session — updating stored phone numbers, toggling verbose logging, or adjusting XMODEM timeouts — because no listener needs to be re-bound.

The main window also has a large **Save and Restart Server** button. Changes that affect how the listeners are bound — turning SSH on or off, changing a listen port, enabling or disabling the serial modem, changing credentials — need a restart to take effect. The button saves the config, sets an internal restart flag, and triggers an orderly shutdown. The main loop detects the flag and re-spawns the server; the GUI reopens after the restart completes.

OUT-OF-BAND CHANGES

If a telnet or SSH session toggles a configuration value (for example the Telnet Gateway's *T* mode toggle, or the in-session Configuration menu), the GUI notices and refreshes its fields — unless you are currently editing them, in which case your edits are not clobbered.

4.4 Live Console Output

The bottom pane of the window shows the server log: new sessions, auth failures, dialup lookups, XMODEM progress when verbose is on, SSH host-key trust decisions, and shutdown messages. It auto-scrolls to the tail and caps at 2000 lines.

5. Telnet Access

The telnet server is on by default and listens on TCP port **2323**. Change the port via the GUI, the in-session *Configuration* menu, or by editing `telnet_port` in `egateway.conf` and restarting.

5.1 Connecting

```
telnet 192.168.1.160 2323
```

Any standard telnet client works: `telnet`, PuTTY, SyncTERM, CGTERM on the Commodore 64, NetSwitcher on CP/M, a hardware WiFi modem plus a real C64, etc. The first thing the server does is negotiate a few telnet options via IAC (`WILL ECHO`, `WILL SGA`, `DO SGA`, `DO TTYPE`, `DO NAWS`), then prompt for terminal detection.

5.2 Terminal Detection

After the initial option handshake you will see:

```
Press BACKSPACE to detect terminal:
```

The byte you send determines the terminal class, as described in chapter 1. If the client already replied with an `IAC SB TTYPE IS ...` subnegotiation during the initial handshake, the terminal is determined from that and the prompt is skipped entirely.

You are then asked to enable colour:

```
Use ANSI color? (Y/N):
```

No default: press ☐ Y or ☐ N.

5.3 Session Limits

- **Concurrent sessions** — capped by `max_sessions` (default 50).
- **Idle timeout** — `idle_timeout_secs` seconds (default 900 = 15 minutes). Set to 0 to disable. Idle timeouts apply to every prompt the server offers, including login, so users on slow modems should raise this.
- **Screen layout** — 40 columns on PETSCII, 80 on ANSI/ASCII, 22 rows of content (the remaining 2 rows are the header and prompt). Enforced by unit tests to avoid layout regressions.

5.4 Network Gating (Unauthenticated Mode)

When `security_enabled` is `false` (the default), the telnet server refuses connections from public IP addresses. Only the RFC 1918 private ranges — `10.0.0.0/8`, `172.16.0.0/12`,

192.168.0.0/16, plus loopback, link-local, IPv6 loopback (::1), IPv6 link-local (fe80::/10), and IPv6 unique local (fd00::/8) — may connect without authentication. Gateway addresses ending in .1 are also rejected so a misconfigured NAT cannot accidentally bridge a telnet session through your router.

To accept connections from any IP address, enable security and configure a strong username and password.

For lab and isolated-network setups where authentication is genuinely unwanted but the gateway still needs to listen on a non-private interface, the `disable_ip_safety` flag drops the allowlist entirely. The GUI Security frame and the in-session telnet menu both gate the toggle behind a confirmation popup that explains the risk before the change sticks; once enabled, *any* source address may connect with no credentials. Treat it as the explicit "I understand this is open" switch, not a default.

5.5 The Main Menu

```
A  AI Chat
B  Simple Browser
C  Configuration
F  File Transfer
G  Serial Gateway
R  Troubleshooting
S  SSH Gateway
T  Telnet Gateway
W  Weather
X  Exit
```

G Serial Gateway appears whenever at least one of the two ports (Port A or Port B) is configured in *console* mode (see chapter 9.13). It opens an A/B picker showing both ports' status, then bridges the telnet/SSH session straight to the chosen port's wire so you can drive a microcontroller, RS-232 device, or other serial console remotely. From a session that came in over a serial port itself the option is hidden. Press `ESC` twice to leave the bridge. Both ports can host their own concurrent bridge — they're fully independent.

5.6 The Configuration Menu

Pressing `C` on the main menu opens a terminal-side configuration menu that mirrors most of what the GUI exposes. It is intended for headless systems and for the SSH-only path where the GUI is unreachable. Changes are written to `egateway.conf` on confirm and apply immediately to running services where possible.

Port A: <Disabled / Modem mode / Console mode>

Port B: <Disabled / Modem mode / Console mode>

E Security

G Gateway Configuration

M Serial Configuration

S Server Configuration

F File Transfer

O Other Settings

R Reset Defaults

Q Back H Help

E Security

Toggle `security_enabled` and edit the unified `username / password` pair shared by telnet, SSH, and the configuration web UI.

G Gateway Configuration

Outbound gateway settings — telnet negotiation flags, SSH-gateway auth mode, browser homepage, weather ZIP, AI key.

M Serial Configuration

Opens an A/B picker submenu listing both ports with their current status. Pick a port to enter that port's settings menu, where you can toggle its mode (per-port **T**: *Toggle Modem/Console mode*), set its device, baud, framing, flow control, AT-state, dialup mapping, and ring emulator. Each port's settings are fully independent and persist under separate `serial_a_*` / `serial_b_*` keys. The per-port **T** item is hidden from sessions that arrived over a serial port (flipping that port to console would tear down the caller's own connection before it could confirm).

S Server Configuration

Listen ports and enable flags for telnet, SSH, the standalone Kermit server, and the configuration web UI; max sessions, idle timeout, and the network-safety opt-out (`disable_ip_safety`).

F File Transfer

Per-protocol timeouts and retries for XMODEM, ZMODEM, Kermit, and Punter, plus the Kermit server toggle / port and `allow_atdt_kermit`.

O Other Settings

Verbose XMODEM logging and the Show-GUI-on-Startup flag.

R Reset Defaults

Restore every recognised key to its built-in default. Confirms with Y/N first; credentials are reset to `admin / changeme`, so you'll want to set new ones immediately afterward if security is enabled.

GUI VS TERMINAL CONFIG

The GUI in chapter 4 is the friendlier interface for desktop systems. The terminal-side menu exists so that an SSH-only deployment, a serial-console session into the gateway, or any other no-display setup still has a way to edit settings without dropping out to a shell to hand-edit `egateway.conf`.

6. SSH Access

The SSH interface is the recommended way to connect to the gateway from outside your trusted LAN. It is *not* enabled by default — it runs on port **2222**, authenticates against the unified `username` / `password` pair shared with telnet and the web UI, and exposes the same menu system as telnet.

6.1 Enabling the SSH Server

Toggle the *SSH* checkbox in the GUI Server frame, or edit `egateway.conf`:

```
ssh_enabled = true
ssh_port    = 2222
username    = admin
password    = pick-something-strong
```

The same `username` / `password` pair is used by telnet, SSH, and the web configuration UI — changing them here changes authentication for all three. Versions prior to v0.6.0 had separate `ssh_username` / `ssh_password` keys; those keys are no longer recognized, but an upgrading config gets a one-time migration: non-default legacy SSH credentials are adopted into the unified pair on the first load when the unified pair is still at the factory defaults (a *Note: migrating legacy ssh_username=...* line lands in the log). After the next save the legacy keys disappear from the file.

Then click *Save and Restart Server*, or restart manually. On first start with SSH enabled, the gateway generates an Ed25519 host key and persists it to `ethernet_ssh_host_key` next to the binary (mode `0600` on Unix). Subsequent starts reuse the same key so clients can verify the server's identity.

6.2 Connecting

```
ssh admin@192.168.1.160 -p 2222
```

The first time your client connects it will display the server's SHA-256 fingerprint and ask whether to trust it. Accept, and the key is recorded in your client's `~/.ssh/known_hosts` as usual.

After authentication you land on the same Ethernet Gateway menu as a telnet user, rendered in ANSI terminal mode. All features — file transfer, SSH gateway, telnet gateway, browser, AI chat, modem emulator, weather — are available.

6.3 Public-Key Authentication

The built-in SSH server supports password authentication only. However, the *outbound* SSH gateway (chapter 10) offers Ed25519 public-key auth to remote hosts — a separate concept.

PLAINTEXT CREDENTIALS

The unified username / password pair in `egateway.conf` is stored in plaintext (used by telnet, SSH, and the web UI). The SSH connection itself is encrypted, but the file is not. Protect it with the `0600` permissions the gateway sets automatically, and do not reuse passwords from other services.

6.4 Web Configuration Interface

The *configuration web server* renders the same settings page the desktop GUI does, in a browser. It is off by default; toggle it on in the GUI's Server frame (the *Web Server* checkbox between Telnet and Kermit Server) or via the telnet menu's **Configuration > Server Configuration > W** key. The default port is **8080**.

```
web_enabled = true
web_port    = 8080
```

Once enabled and the server is restarted, browse to `http://<server-ip>:8080/`. The page mirrors every GUI frame — Server, Security, File Transfer, AI/Browser/Weather, Serial Ports, General — and each frame has its own Save button matching the GUI's semantics:

- **Server: Save and Restart** — persists every field and cycles the gateway through a full restart so new listener ports and enable-flags take effect (same flow as the GUI's *Save and Restart* button).
- **Serial Ports: Save** — persists every field and reopens the serial port managers only; telnet, SSH, and the web UI itself stay running.
- **Security / File Transfer / AI/Browser / General: Save** — persists only. The relevant code paths read these settings live, so no restart is needed.

The page also includes a live `Console Output` tail (polled every two seconds), a serial-port dropdown populated from the same device enumeration the GUI uses (with a refresh button that re-scans without a page reload), and inline confirmation dialogs that warn the operator before disabling the web server or changing its port — either action breaks the currently-open browser session, so the page nudges the operator to reconnect at the new address after saving.

6.4.1 Authentication and Network Gating

The web server respects the same `security_enabled` / `disable_ip_safety` flags telnet does:

- When `security_enabled` is **off** (the default), access is gated on source IP — only private / loopback / link-local addresses are accepted, and gateway-style `*.*.*.1` addresses are rejected. No password is required.
- When `security_enabled` is **on**, the web server prompts via HTTP Basic Auth against the unified username / password pair (the same one telnet and SSH use). The IP-safety allowlist is bypassed because the password is now the gate.
- Three failed Basic-Auth attempts from the same source IP trip the shared 5-minute lockout used by telnet and SSH. Subsequent requests get HTTP 429 Too Many Requests with a

`Retry-After: 300` header until the window expires. The lock fires *before* request parsing, so a flood of malformed POSTs from a banned IP can't keep the server busy either.

6.4.2 Defense-in-Depth Caps

- **30-second read timeout** per request — a slow-loris client can't park a worker indefinitely.
- **16 concurrent connections max**; excess receive `HTTP 503 Service Unavailable` with a short `Retry-After`.
- 16 KB cap on request headers, 64 KB cap on POST bodies — the save form is far smaller; this just bounds the worst case from a hostile peer.

ENCRYPTION

The configuration web server speaks plain HTTP, not HTTPS. Don't expose it directly to the public Internet — use it on a trusted LAN, or tunnel via SSH (`ssh -L 8080:127.0.0.1:8080 ...`) when configuring the gateway from elsewhere.

7. Security

Security is optional. In the default configuration the gateway trusts anyone on your private LAN and refuses public-Internet connections. Turn security on when you want explicit authentication, or when you need to accept connections from public addresses over SSH.

7.1 Enabling Login

Check the *Require Login* box in the GUI Security frame, or set:

```
security_enabled = true
username = admin
password = your-password
```

A single `username` / `password` pair gates all three authenticated interfaces — telnet, SSH, and the configuration web UI. Three failed logins from the same source IP — across any of the three — trip a 5-minute lockout that blocks every protocol on that IP.

7.2 Credential Comparison

Passwords are compared byte-for-byte using a constant-time equality check (`constant_time_eq`) so an attacker cannot measure tiny timing differences to recover the password one character at a time. This matters less over telnet (because the password is transmitted in cleartext anyway) but it is a genuine defence for SSH logins.

7.3 Per-IP Lockout

The telnet, SSH, and web servers share a single in-memory lockout table keyed on the remote IP address. After **3** failed authentication attempts within 5 minutes the IP is blocked across *all three* protocols for 5 minutes. An attacker cannot bounce between protocols to reset the counter. The web server responds to a locked-out client with `429 Too Many Requests` and a `Retry-After: 300` header; telnet and SSH simply refuse the session.

Once the block expires the counter resets on the next successful login. Lockout events are logged to the console panel so a legitimate user who locked themselves out can see what happened.

7.4 File Permissions

On Unix, every file the gateway writes that may contain secrets is set to mode `0600` (owner read/write, no access for group or other):

- `egateway.conf` — the unified login password (used by telnet, SSH, and the web UI), Groq API key.
- `dialup.conf` — no secrets, but dial-up mappings.
- `gateway_hosts` — known host fingerprints for outbound SSH dials.

- `ethernet_ssh_host_key` — the SSH server's private host key.
- `ethernet_gateway_ssh_key` — the gateway's outbound SSH client key.

Config file writes are done to a per-PID temporary filename and then renamed into place. This closes a TOCTOU window against symlink-swap attacks in shared working directories.

7.5 Audit Logging

Important events are written to the live console panel (and to stderr in headless mode):

- Auth successes and failures, with the remote IP.
- Lockout events when an IP hits the 3-failure threshold.
- SSH Gateway host-key trust decisions — TOFU accept, key update, and rejection — with host, port, key algorithm, and SHA-256 fingerprint. This gives you a forensic trail if a changed host key turns out to be a man-in-the-middle attempt.

8. File Transfer

File transfer is the original reason this project exists. The gateway implements six protocols, covering everything from a 1977 vintage client to a modern terminal that drag-and-drops files onto the window:

XMODEM (classic, 128-byte blocks)

The original 1977 protocol. Uses a `S0H` header, a block number and its complement, 128 bytes of payload, and either an 8-bit checksum or a CRC-16. Compatible with every retro terminal program ever made.

XMODEM-1K (1024-byte blocks)

A 1980s extension that uses an `STX` header and 1024 bytes of payload. Reduces protocol overhead by a factor of eight. Senders may mix 1K and 128-byte blocks freely within a single transfer; the gateway branches on the header byte per block.

YMODEM (batch + metadata)

XMODEM-1K plus a block-0 header. Block 0 has `S0H`, block number `0`, complement `0xFF`, and a 128-byte payload containing the filename, size, and (optionally) mtime. The receiver uses the size to truncate trailing `SUB` padding exactly.

ZMODEM (Forsberg 1988)

Streaming protocol with ZDLE escaping, hex / binary16 / binary32 headers, CRC-16 and CRC-32, batch send and receive, ZSKIP, and an autostart trigger so the user can drop a file onto the terminal from any menu without first navigating to the file-transfer screen. See §8.5.

Kermit (Columbia, multi-flavor)

The classic over-7-bit-line protocol with sliding windows, long packets, attribute packets, and resume. The gateway acts as both client (push/pull from the menu) and server (idle in `kermit-server` mode for incoming GET / SEND / FINISH commands), and accepts a standalone TCP listener as well. See §8.8.

Punter C1 (“New Punter”, Commodore)

The single-file protocol that CCGMS, Novaterm, and StrikeTerm speak natively on Commodore 64/128 BBSes. A two-phase handshake (a file-type block, then the data blocks) with two per-block checksums (a 16-bit additive checksum and a 16-bit cyclic checksum) and 3-byte ASCII handshake codes (GOO / BAD / ACK / S/B / SYN). This is single-file C1 — not the older PET “PTP” or multi-file “MPP.” See §8.9.

8.1 The Transfer Directory

All transfers are scoped to the `transfer_dir` setting (default `transfer`), relative to the gateway's working directory. The gateway canonicalises the path on every access to prevent symlinks from escaping the jail, and filenames are validated to reject path traversal.

Filename rules:

- Letters, digits, dots, hyphens, and underscores only.

- Maximum 64 characters.
- Cannot start with a dot.
- Cannot contain ...
- Must include at least one letter or digit.

File size is capped at **8 MB**. Larger uploads are aborted server-side with three `CAN` bytes.

8.2 Navigation

From the main menu press `F`:

```

U Upload          (receive file from your terminal)
D Download        (send file to your terminal)
X Delete          (remove a file from the transfer dir)
C Change directory (subdirectory within transfer_dir)
K Kermit server mode (idle and wait for client commands)
I Toggle IAC escaping (for binary files with 0xFF bytes)

```

8.3 Uploads — Protocol Selection & Auto-Detection

Press `U` and enter a filename. Before the transfer starts, the gateway asks which protocol to use:

SELECT UPLOAD PROTOCOL

```

X XMODEM/YMODEM 128/1K, auto
Z ZMODEM         1K, autostart
K KERMIT         any flavor, auto
P PUNTER         C1 CCGMS/Novaterm

```

Pick one, or ESC to cancel:

The choice is immediate — one key, no Enter required. ESC (or `←` on PETSCII, byte 0x5F) cancels. The gateway then prints the matching *Start ... send* prompt and waits for your terminal to begin. For the XMODEM/YMODEM family it prints:

```
Start XMODEM/YMODEM send from your terminal now.
```

The four families differ in how the incoming transfer is handled:

- **XMODEM/YMODEM** (key `X` or `Y`) — accepts classic XMODEM (128-byte `SOH`), XMODEM-1K (1024-byte `STX`), or YMODEM (block-0 filename/size header) and *auto-detects which one the sender is using*; the logic is detailed below.
- **ZMODEM** (key `Z`) — receives a ZMODEM stream, including a multi-file batch (colliding names are declined via ZSKIP).

- **Kermit** (key **K**) — receives any Kermit flavor; the peer's capabilities are read from its Send-Init.
- **Punter** (key **P**) — receives a single-file Punter C1 transfer (§8.9). The file is saved under the name you entered, with a `.prg/.seq/.usr` extension appended to match the sender-declared CBM type when your filename has none.

Within the **XMODEM/YMODEM** family the gateway auto-detects the exact variant. The logic is:

1. The receiver first sends `'C'` (0x43) for CRC-16 mode. It retries every three seconds for two thirds of the negotiation timeout. Most modern clients respond to this.
2. If no response comes, the receiver falls back to sending `NAK` (0x15) to request checksum mode. This rescues very old clients that never implemented CRC.
3. The first header byte from the sender determines the block size: `S0H` (0x01) means a 128-byte block, `STX` (0x02) means a 1024-byte XMODEM-1K block.
4. If the first header is `S0H` with block number `0` and complement `0xFF`, it is a **YMODEM block 0**. The gateway reads the filename header, parses the file size, validates the CRC, ACKs, and sends a second `'C'` to start the data phase.
5. Within a single transfer the sender may mix 128-byte and 1024-byte blocks freely. Each block's header byte selects the payload length.
6. Transfer ends on `EOT` (0x04), which the server ACKs. For YMODEM, the server then sends `'C'` and consumes the end-of-batch "null block 0" the sender emits to signal "no more files" (Forsberg §7.4).
7. Once the transfer is complete, the file is truncated: YMODEM uses the exact size field from block 0 so files that legitimately end in `0x1A` bytes (EXEs, some archives) aren't corrupted; plain XMODEM and XMODEM-1K strip trailing `SUB` (0x1A) padding since there's no length metadata.

Cancellations are handled both ways: sending `ESC` from your side aborts the transfer, and three `CAN` bytes from either side abort with "Transfer cancelled".

DUPLICATE FILENAME

If the name you type is already in the transfer directory, the server asks `Overwrite? (Y/N)` *before* starting the transfer. Answering ☐ `N` cancels cleanly; answering ☐ `Y` proceeds and truncates the existing file on save. This prevents a whole transfer from running only to fail at the final write step.

8.4 Downloads — Protocol Selection Prompt

Press ☐ `D`, then pick a file from the numbered list (paginated, 10 per page). Before the transfer starts, the gateway asks which protocol to use:

SELECT PROTOCOL

X	XMODEM	128-byte blocks
1	XMODEM-1K	1024-byte blocks
Y	YMODEM	name+size hdr, 1K
Z	ZMODEM	autostart, 1K
K	KERMIT	any flavor, auto
P	PUNTER	C1 CCGMS/Novaterm

Pick one, or ESC to cancel:

Pick the protocol that your terminal program supports. The choice is immediate — one key, no Enter required. ESC (or on PETSCII, byte 0x5F) cancels.

The selection affects what the sender does:

- **XMODEM** — the sender transmits 128-byte `S0H` blocks exclusively.
- **XMODEM-1K** — the sender transmits 1024-byte `STX` blocks, with an opportunistic 128-byte `S0H` fallback on the final short block.
- **YMODEM** — the sender begins with a block-0 `S0H 0x00 0xFF` header containing the filename and size, waits for `'C'` acknowledgement, then transmits 1K data blocks just like XMODEM-1K.
- **ZMODEM** — the sender emits a `ZRQINIT` hex header, then negotiates flags via `ZRINIT` / `ZSINIT`, sends a `ZFILE` subpacket with filename + size + mtime + mode, streams `ZDATA` with CRC-16 or CRC-32 subpackets, and finishes with `ZE0F` + `ZFIN`. `ZSKIP` allows the receiver to decline individual files in a batch.
- **Kermit** — the sender first exchanges `S/I` negotiation packets (block-check type, packet length, window size, capabilities), then sends one or more `F/D/Z` file groups with optional `A` attribute packets for size/mtime/mode/resume, and ends the batch with `B`. Long packets and sliding windows are negotiated based on what the peer offers.
- **Punter** — the gateway first sends a file-type block (PRG, SEQ, or USR), then streams the data blocks, each carrying a 248-byte payload behind a 7-byte header and protected by both a 16-bit additive checksum and a 16-bit cyclic checksum. Handshaking is driven by the 3-byte ASCII codes `GOO` / `BAD` / `ACK` / `S/B` / `SYN`. The outbound file type is auto-detected from the filename extension (text-like `.seq` / `.txt` / `.doc` → `SEQ`, `.usr` → `USR`, else `PRG`).

8.5 ZMODEM Autostart Detection

ZMODEM (Chuck Forsberg's 1988 protocol) is fully supported, including batch send, batch receive, and ZSKIP. Many terminal programs (Qodem, ZOC, SyncTerm, ExtraPuTTY) start a ZMODEM send by emitting the autostart sequence `**\x18B`, `**\x18A`, or `**\x18C` directly into whatever menu the user happens to be sitting at — users expect to drag a file onto the

terminal and have it transfer without first navigating to a menu. The gateway's idle input loop watches for that prefix and:

1. Logs "File transfer: ZMODEM autostart detected; switching to receive".
2. Drains any residual ZRQINIT bytes the sender already pushed (it'll retransmit once we ZRINIT).
3. Hands off to the regular ZMODEM receive flow — the sender's filename is used (after path-traversal validation), with mtime and mode applied if carried in the ZFILE subpacket.
4. Returns to whichever menu the user was on when the autostart fired.

You can also drive a ZMODEM transfer manually: from the File Transfer menu pick **U** (upload) or **D** (download), select **Z** at the protocol prompt, and start the send/receive on your terminal. Manual mode is the only option for clients that don't emit the autostart prefix (some MS-DOS-era implementations, scripted `sz` on Unix).

ZCOMMAND IS RECOGNIZED BUT NEVER EXECUTED

The ZMODEM spec defines one *optional* frame the gateway deliberately does **not** implement: ZCOMMAND (frame 0x12). It is not part of ZMODEM's required core — a fully spec-compliant implementation need not support it (Forsberg's own reference receiver, `rz`, refuses it unless started with an explicit `--allow-commands` flag), and most transfers never use it. It lets a connected *sender* push a shell command for the *receiver* to run, returning the command's exit status in a ZCOMPL header — on Chuck Forsberg's original Unix `rzsz` reference code the command string is handed straight to `/bin/sh -c` (its classic use was a sender pushing `sz somefile` to auto-start a download). That is, in plain terms, **arbitrary remote code execution on the host**, which is an unacceptable default for a long-lived, multi-session server: unlike Forsberg's throwaway single-session `rz`, one ZMODEM session here shares the process with every other connected user. Rather than ignore the frame, the gateway handles it defensively — it recognizes ZCOMMAND, drains the command subpacket (so the wire stays in sync), replies a non-zero ZCOMPL (refused), and ends the session. Every ZCOMMAND request is rejected, and there is no configuration switch to enable it. If you need a shell on the gateway host, use the authenticated [SSH interface](#) (Chapter 6) — that is the proper, encrypted, login-gated path for command access. Every other Forsberg 1988 frame is fully implemented.

8.6 Telnet NVT Transforms

Telnet's Network Virtual Terminal (RFC 854) defines two wire-level transforms for a text stream. For binary file transfers the gateway applies only the first and treats the channel as 8-bit transparent (RFC 856 binary-transmission semantics):

- **IAC escaping (applied).** Byte 0xFF is the IAC (Interpret As Command) marker. Any 0xFF in data — including a CRC byte that happens to be 0xFF — must be doubled on the wire as 0xFF 0xFF. The receiver collapses the pair back to a single 0xFF.
- **CR-NUL stuffing (NOT applied).** RFC 854 says a bare 0x0D (CR) in NVT *text* should be sent as CR NUL (0x0D 0x00). The gateway does **not** do this for file transfers: the payloads are binary, and inserting a 0x00 after every 0x0D corrupts the data for any peer that doesn't strip it back out — real C64/CP-M clients and telnet↔serial bridges (e.g. `tcpser`) do not. CR and every other non-IAC byte pass through literally.

The honest reality: there's no way for the server to know for certain whether a given client actually expects IAC escaping during an XMODEM/YMODEM transfer. Strict clients (PuTTY / ExtraPuTTY, Tera Term, C-Kermit, SecureCRT) do; many retro clients (IMP8, CCGMS, StrikeTerm, NovaTerm, some AltairDuino firmware) don't — they treat a TCP connection to port 23 as raw bytes. Rather than guess from the terminal type, the gateway keys the default on a signal it can actually observe: *whether the client engaged in telnet option negotiation during the opening handshake*. A client that answered our `WILL/DO` options (the strict clients above) gets IAC escaping **on**; a client that stayed silent — raw-TCP retro clients, and anything on a serial line, which skips negotiation entirely — gets it **off**. That observed behaviour correlates closely with the detected terminal type, so in practice the defaults line up like this:

Terminal (typical)	Typical client	Negotiates?	Auto default
ANSI	PuTTY / ExtraPuTTY, Tera Term, C-Kermit, SecureCRT, xterm	yes	IAC on
PETSCII	Commodore 64/128 (IMP8, CCGMS, StrikeTerm, NovaTerm)	no (raw TCP)	IAC off
ASCII	Dumb terminals, AltairDuino, custom hardware	no (raw TCP)	IAC off

SSH and serial sessions skip both transforms unconditionally — SSH has no in-band command byte, and serial streams aren't subject to NVT rules.

If the auto default is wrong for your client — the most visible symptom is consistent block-CRC failures in one direction — press ☐ on the File Transfer menu to flip both transforms and retry. The choice is session-local; it isn't persisted to `egateway.conf`, so the next connection from the same client gets the same auto default again (the client negotiates — or stays silent — the same way each time).

WHAT YOU'LL SEE IF NVT IS WRONG

An upload that runs fine for one or two dozen blocks and then CRC-fails repeatedly is usually the first block containing a raw `0xFF` or `0x0D` byte that isn't being escaped by one side. A download that stalls after 3–4 blocks with the client repeatedly sending 'C' instead of ACK is the same thing on the send side. Toggle ☐ and retry.

8.7 Protocol Timeout Tunables

Transfer timings are grouped by protocol family. The *XMODEM-family* keys apply to classic XMODEM, XMODEM-1K, and YMODEM (which share a single send/receive code path in `xmodem.rs`). ZMODEM, Kermit, and Punter each have their own independent sets because their frame- or block-based protocols have different cadence requirements.

Each key can be changed from the telnet *Configuration* → *F File Transfer* → *X / Y / Z / K / P* menus, from the GUI *File Transfer* → *More...* popup, or by editing `egateway.conf` directly. Changes take effect on the next file transfer — no restart required.

XMODEM family (XMODEM / XMODEM-1K / YMODEM)

Key	Default	Meaning
xmodem_negotiation_timeout	45 s	Maximum time the receiver will wait for the first SOH/STX header from the sender, and vice versa.
xmodem_negotiation_retry_interval	7 s	Gap between successive C/NAK pokes during the initial handshake. Christensen's original XMODEM and Forsberg's reference implementations specify ~10 s; 7 s is a compromise that starts quickly but avoids stockpiling extras in a slow-starting sender's buffer (which can be misinterpreted as "retry block 0" requests, causing spurious duplicate-block retransmissions).
xmodem_block_timeout	20 s	Per-block timeout. If a block header does not arrive within this long after an ACK/NAK, the receiver retries.
xmodem_max_retries	10	Maximum NAK retries per block before the receiver cancels the transfer with three CANs.

ZMODEM

Key	Default	Meaning
zmodem_negotiation_timeout	45 s	Maximum time to complete the ZRQINIT / ZRINIT handshake before giving up on the session.
zmodem_negotiation_retry_interval	5 s	Gap between ZRINIT / ZRQINIT re-sends during the handshake. Analogous to the XMODEM retry interval but for ZMODEM's frame-based protocol.
zmodem_frame_timeout	30 s	Per-frame read timeout once a transfer is live. Applies to headers and subpackets alike.
zmodem_max_retries	10	Retry cap for ZRQINIT, ZRPOS, and ZDATA frames. Once exceeded, the session cancels with a ZABORT.

Punter (C1)

Key	Default	Meaning
punter_block_size	255	Maximum block size in bytes (8-255). A block carries a 248-byte payload behind a 7-byte header at the default. Lower it toward ~40 on noisy lines to shrink the retransmit unit.
punter_negotiation_timeout	45 s	Maximum time to complete the file-type handshake before giving up on the session.
punter_negotiation_retry_interval	5 s	Gap between handshake re-sends while waiting for the peer to respond.
punter_block_timeout	20 s	Per-block read timeout once a transfer is live.
punter_max_retries	10	Retry cap per block (BAD/SYN re-sends) before the session aborts.
punter_max_bad_rounds	30	Consecutive corrupt-block resend rounds tolerated before giving up — kept higher than punter_max_retries because a real C64 peer never caps its resends, so a low value would make the gateway quit first and strand the peer.
punter_hangup_on_failure	false	On give-up, drop the connection (carrier) so a C64 — which C1 cannot be told to abort — sees loss-of-carrier and exits instead of hanging. Ends the whole session; off by default.

VERBOSE DIAGNOSTICS

Set `verbose = true` to log every header byte, CRC result, block number, and timeout event. This is invaluable when a transfer fails on a specific block and you need to know which side hiccuped. The log identifies each event with its protocol prefix (XMODEM recv: , ZMODEM send: , etc.) so multi-protocol sessions remain traceable.

8.8 Kermit — Server & Client Mode

Beyond XMODEM/YMODEM/ZMODEM, the gateway implements a spec-complete Kermit file-transfer protocol following Frank da Cruz's *Kermit, A File Transfer Protocol* (Digital Press, 1987) plus the long-packet, sliding-window, and streaming extensions used by C-Kermit. Both directions are supported: the gateway can act as a Kermit *server* (idle waiting for commands from a connecting client) and as a Kermit *client* (connect to a remote server and pull/push files programmatically).

Why Kermit alongside XMODEM/ZMODEM?

Kermit was designed in 1981 for environments where bytes can be lost, mangled, or stripped to 7 bits — old serial links, packet-switched networks, and bridged terminal hardware. It survives conditions that XMODEM/YMODEM/ZMODEM cannot: 7-bit-only links (mainframe gateways, some telnet bridges), lossy connections that need selective retransmit, and multi-

platform interop (every retro platform that mattered has a Kermit). ZMODEM is faster on clean modern links; Kermit is the right choice for the gnarly ones.

Entering server mode

There are three entry paths into Kermit Server Mode:

- **Telnet / SSH menu** — from the File Transfer menu press `[K]`. Honors authentication, the network gating allowlist, and the per-session sandbox the same as any other gateway feature.
- **Standalone TCP listener** — set `kermit_server_enabled = true` and the gateway binds `kermit_server_port` (default 2424). Every accepted connection drops straight into protocol mode — no menu, no auth gate. Use it from C-Kermit with `kermit -j host:2424`.
- **Serial dial** — with `allow_atdt_kermit = true`, dialing `ATDT KERMIT` from the modem emulator hands the serial caller to the Kermit dispatcher directly. See section 9.2.

The bottom two paths bypass `security_enabled` by design — they exist so vintage receivers that can't paint a menu still work. Both are off by default and have explicit GUI / telnet warnings before they enable.

However the session enters server mode, the terminal displays a banner listing the supported peer commands (`send`, `get`, `remote dir`, `remote cwd`, `remote type`, `remote delete`, `remote rename`, `remote mkdir`, `remote rmdir`, `remote kermit`, `remote help`, `finish`, `bye`). The gateway then idles waiting for protocol packets; when the peer sends `finish`, `bye`, or disconnects, control returns to whatever surface launched the session: the File Transfer menu (telnet/SSH path), a closed TCP socket (standalone listener), or `NO CARRIER` (ATDT KERMIT). The File Transfer path also shows a per-file ✓/× summary of received files.

What the peer can do

Peer command	Wire packet	Effect
send <file>	S + F + A + D... + Z + B	Upload a file from the peer to the gateway's transfer_dir (or current sub-directory).
get <file>	R, then we send	Download a file from the gateway to the peer.
remote dir	G D	List files in the current effective directory.
remote cd <subdir>	G C	Change the gateway's working sub-directory for subsequent commands. Field-encoded path argument per spec §6.7. Bare remote cd (no argument) resets to the transfer_dir root; remote cd .. (cdup) pops one component (no-op at root); a target that doesn't exist on disk is refused with E-packet rather than silently ACKed.
remote space	G \$	Report free disk bytes at the current effective directory.
remote kermit	G K	Identify the server (Ethernet Gateway Kermit 0.6.3).
remote help	G H / G ?	List supported subcommands.
remote delete <file> / remote era	G E	Delete a file in the current effective directory. Path-traversal rejected; missing files return E-packet (not silent ACK). One filename per call — no wildcard expansion.
remote rename <old> <new>	G R	Rename a file in place. Both names field-encoded per spec §6.7. Server refuses if the destination already exists, to avoid silent clobber.
remote type <file>	G T	Display a file's contents on the peer's terminal via the X+Z inverse-transfer pattern. Intended for text files; binaries will spew control bytes.
remote mkdir <dir>	G m	Create an empty subdirectory under the current effective directory. Wire letter is <i>lowercase</i> m (per C-Kermit's setgen('m', ...)) — distinct from M reserved for remote message.
remote rmdir <dir>	G d	Remove an <i>empty</i> directory. Lowercase d wire letter — distinct from D (DIRectory listing). Non-empty directories are refused; clear contents first with remote delete.
finish / bye / logout / remote exit	G F / G L / G I / G X	End the protocol session politely. The gateway accepts all four since C-Kermit emits different letters for them (e.g. logout sends G I, remote exit sends G X).
<i>any other</i> remote ...	G letter	Refused with an explicit E-packet (per spec §6) so the peer's UI surfaces a clean “command not supported” error instead of a phantom-success ACK.

Capabilities advertised

The gateway negotiates the full set of modern Kermit extensions: long packets (up to 9024 bytes per packet), sliding windows (up to 31 outstanding packets with selective-repeat retransmit), streaming (no per-packet ACKs on reliable links), CRC-16 block check, attribute packets carrying file size + mtime + mode + record format + creator ID + ~10 other typed metadata fields, repeat-count compression, RESEND (resume partial uploads), and locking shifts (SO/SI region quoting for 8-bit transit on 7-bit links).

Two of these are off by default and gated behind config keys: `kermit_resume_partial` for RESEND (off because a peer that doesn't honor `disposition='R'` would corrupt the result), and `kermit_locking_shifts` for SO/SI quoting (off because no modern peer uses it — QBIN per-byte 8th-bit prefixing covers the same use case).

C-Kermit interop

Real C-Kermit 10.0 interop is verified by the test suite for both directions. Example invocations:

```
$ kermit -B -j 192.168.1.160:2323 -i -q -s ./localfile.bin
# Upload localfile.bin to the gateway. -B batch, -j telnet-protocol TCP,
# -i image (binary), -q quiet, -s send action.

$ kermit -B -j 192.168.1.160:2323 -i -q -g report.txt -a /tmp/report.txt
# Pull report.txt from the gateway and save locally as /tmp/report.txt.
# -g get action, -a alternate local name. The gateway must be in
# Kermit Server Mode (File Transfer menu → K) when ckermit dials in.
```

FULL KERMIT REFERENCE

A standalone reference page [kermit.html](#) documents every G subcommand, the full capability table, all 18 `kermit_*` configuration keys with defaults (including `kermit_idle_timeout` which accepts 0 to disable the server-mode disconnect), additional examples, and spec citations. See it on the project website alongside this manual.

8.9 Punter (C1) — Commodore

Punter is the file-transfer protocol that Commodore 64/128 terminal programs — CCGMS, Novaterm, and StrikeTerm — speak natively on the BBSes of that era. The gateway implements single-file **C1** (“New Punter”): not the older PET PTP and not the multi-file MPP variant. It works over both the telnet path (with IAC escaping) and the serial-modem path, and shares the same 8 MB maximum file size as the other protocols.

How a C1 transfer runs

C1 is a stop-and-wait, block-at-a-time protocol coordinated entirely by 3-byte ASCII handshake codes:

Code	Meaning
G00	Block received intact; send the next one.
BAD	Checksum mismatch; retransmit the last block.
ACK	Acknowledge a control transition.
S/B	“Send/Begin” — ready to receive the next block.
SYN	Re-synchronise after a stall or a lost block.

1. **Two-phase handshake.** The transfer opens with a file-type block, then proceeds to the data blocks. Each block is protected by *two* independent 16-bit checksums — an additive checksum and a cyclic checksum — so a block is only accepted (G00) when both agree; otherwise the receiver answers BAD and the sender retransmits.
2. **Block layout.** Blocks are up to 255 bytes — a 248-byte payload plus a 7-byte header. The block size is configurable from 8-255 (default 255) via punter_block_size ; lower it toward ~40 on a noisy line so each retransmit costs less.
3. **File-type detection on download.** When sending a file to the C64, the outbound file type is auto-detected from the filename: text-like extensions (.seq , .txt , .doc) are sent as SEQ , .usr as USR , and everything else as PRG .

Selecting Punter

Punter appears in both the upload and download protocol pickers under the key P (“PUNTER”). From the File Transfer menu, press U or D , choose your file, and pick P at the protocol prompt; then start the matching transfer in your Commodore terminal. The per-protocol tunables (block size, negotiation timeout, retry interval, block timeout, max retries) live on the *Configuration* → *F File Transfer* → *P Punter* screen, in the GUI's File Transfer *More...* popup, and on the web config page — see §8.7 for the punter_* key reference.

CANCELLING AND RESTARTING

Punter has no in-band cancel signal — when you abort a transfer on the Commodore (RUN/STOP in CCGMS), the terminal stops locally and sends *nothing* over the wire, so the gateway cannot tell you bailed. It keeps the old transfer alive until it times out (up to punter_block_timeout , default 20 s), then prints *Transfer failed* and returns to the menu.

Wait for that Transfer failed message before restarting. If you re-arm your Commodore receiver while the gateway is still finishing the aborted transfer, its fresh handshake collides with the dying one and the two ends desync. The reliable sequence is: cancel on the Commodore → wait for the gateway to report *Transfer failed* and redraw the menu → start the new transfer on the gateway → then start the matching transfer on the Commodore. (To shorten the wait, lower punter_block_timeout .) The gateway clears any stray bytes the aborted transfer left in the line before each new Punter transfer, so a clean restart resyncs reliably.

9. Serial / Hayes AT Modem Emulation

The modem emulator turns the gateway into a virtual Hayes Smartmodem on a physical serial port. Retro hardware — a Commodore 64 with an RS-232 cartridge, a CP/M machine, an AltairDuino 2SIO2, an RC2014 — connects as if the gateway were a real modem. `ATDT`, `ATH`, `+++`, all of it works.

The gateway exposes **two physically independent serial ports, Port A and Port B**, each with its own enabled flag, mode, device, baud, AT/S-register state, and stored phone-number slots. The two ports run in separate manager threads and persist under separate `serial_a_*` / `serial_b_*` key blocks, so you can run a Hayes modem on one wire and a telnet-serial bridge on the other (or two modems, or two bridges) simultaneously without any cross-talk.

The emulator speaks a full Hayes command set on each port: `AT`, `ATZ`, `AT&F`, `AT&W`, `AT&V`, `ATE`, `ATV`, `ATQ`, `ATI` variants, `ATH`, `ATA`, `ATO`, `ATX`, `AT&C`, `AT&D`, `AT&K`, `AT&Z`, `AT+PETSII`, `ATD` (with `T`, `P`, `L`, `S` variants), `ATSn`, S-registers S0-S26, the `A/` repeat-last-command shortcut, and the `+++` escape sequence with S2/S12 guard-time semantics. `AT&W` on Port A persists into the `serial_a_*` keys and never touches Port B's state, and vice versa.

9.1 Enabling a Port

Each port enables and configures independently. From the GUI Serial Port frame:

1. Check the *Enabled* box for the port you want to bring up — Port A and Port B each have their own checkbox in the frame's header row.
2. Pick a device from that port's dropdown. If your port does not appear, plug in the adapter and click the refresh (↻) button on Port A's row to re-scan both lists.
3. Set the baud rate inline on the port's row.
4. Click that port's *More...* button to choose *Mode* (Modem vs. Telnet-Serial), data bits, parity, stop bits, flow control, and the full Hayes AT state.
5. Click the *Save* button on the frame's header row. Both ports' settings are persisted atomically and each manager reconfigures with the new values.

You can also configure everything live from an in-session telnet / SSH user: *Configuration* → *M Serial Configuration* → *A* or *B*. That per-port menu has ☐ `E` to enable, ☐ `T` to toggle this port between modem and console mode, ☐ `S` to detect and select a device, and submenus for baud / framing / flow control. Changes applied through the in-session menu take effect immediately on the chosen port without affecting the other port.

SERIAL LOCKOUT PROTECTION

When you change parameters from a serial session (i.e. through the modem itself), the server asks for confirmation and *reverts* if you don't respond within 60 seconds. If your new baud rate is wrong, your serial terminal will stop echoing and you will simply wait out the timer — your old settings come back unchanged.

Typical C64-friendly parameters: 2400–9600 baud, 8N1, no flow control. CP/M systems usually want 9600 8N1. The AltairDuino 2SIO2 is commonly configured for 9600 8N1 as well; remember to press *stop* and *aux1 up* on the front panel to configure its serial ports.

COMMODORE 64 WITH A BIT-BANGED USERPORT SERIAL ADAPTER

If your C64 reaches the gateway through a **bit-banged userport RS-232 adapter** — such as the **EZ232**, which is a passive level-shifter with no UART of its own — use **1200 baud**, not 2400. On these adapters the C64 times every serial bit in software, and it has almost no timing margin to spare. When a long command wraps past column 40 the screen scrolls, the KERNAL briefly masks interrupts to shuffle screen memory, and any keystroke being transmitted during that scroll has its bit timing disrupted. The result is an intermittent, single-character corruption (a typed letter arriving as a stray PETSCII graphics byte) that runs a garbled command. This has been observed with **CCGMS** in PETSCII mode. Dropping both the gateway port (`serial_a_baud = 1200`) and the C64 terminal program to 1200 baud roughly doubles the per-bit timing budget and clears it up. A hardware-UART interface (SwiftLink / Turbo232) or a WiFi modem with its own UART is not affected and can run faster. See §16.9.

DEVICE NAMES CAN MOVE — PORT SETTINGS DON'T

A port's *settings* (mode, baud, framing, AT/S-register state) are keyed to **Port A** / **Port B** and persist across restarts. The *device path* (`serial_a_port` / `serial_b_port`, e.g. `/dev/ttyUSB0`) is just a saved string — and on Linux the kernel hands out `ttyUSB0`, `ttyUSB1`, ... in the order adapters enumerate. After a reboot, or if you unplug the adapters and plug them back in a different order, those names can **swap**, so Port A may suddenly point at the adapter you meant for Port B (with Port A's baud/mode applied to the wrong device). To pin a device regardless of enumeration order, set the port to a stable by-id path instead of `ttyUSBn` — e.g.

`serial_a_port = /dev/serial/by-id/usb-FTDI_FT232R_USB_UART_A1234-if00-port0` (list them with `ls -l /dev/serial/by-id/`). After any re-cabling, confirm each port's device in the GUI dropdown or the telnet *Configuration* → *M* screen.

9.2 Dialing

The emulator routes dial strings over TCP. There is no real phone line — the "call" is a `tokio::net::TcpStream::connect()`.

Command	Action
ATDT host:port	Connect to a remote telnet server. ATDT and ATDP are equivalent (there is no audible difference over TCP).
ATDT ethernet- gateway	Built-in shortcut — dials this gateway's own main menu, so the serial device can use all gateway features (file transfer, browser, AI chat, etc.) without a network detour.
ATDT KERMIT	Drop directly into Kermit server mode — no menu, no banner, indistinguishable on the wire from a real <code>kermit -j host</code> in server mode. Aliases: <code>ATDT kermit</code> , <code>ATDT kermit-server</code> , <code>ATDT kermit server</code> . Requires <code>allow_atdt_kermit = true</code> ; off by default because it bypasses the telnet menu's auth gate. Idle timeout still honors <code>kermit_idle_timeout</code> ; on session end the modem emits <code>NO CARRIER</code> .
ATDP host:port	Same as ATDT — pulse selector, ignored.
ATDSn (n = 0-3)	Dial stored number in slot <i>n</i> . Bare ATDS dials slot 0.
ATDL	Redial the last number.
AT&Zn=s	Store phone number or <code>host:port</code> in slot <i>n</i> (0-3). Preserves hostname case.

Dial-string modifiers (only honoured when the dial string looks like a phone number, i.e. digits and formatting characters):

Modifier	Effect
,	Pause for S8 seconds (default 2) before continuing. Each comma adds S8 seconds.
W	Wait for dial tone. Adds S6 seconds to the pre-delay.
;	After "connect", return to command mode instead of entering online data mode. Applies to hostnames too.
*, #	DTMF digits. Preserved in the dial target for phone-number lookup.
P, T, @, !	Pulse, tone, quiet-answer, hookflash selectors. Accepted, ignored.

Modifiers are *not* stripped from hostnames. Dialling `ATDT pine.example.com` connects to "pine.example.com" verbatim — the heuristic notices that the dial string contains letters and dots and skips modifier processing for everything except the trailing `;`.

9.2.1 Examples

```
ATDT ethernet-gateway
CONNECT 9600
(local gateway main menu appears)

ATDT 192.168.1.160:2323
CONNECT 9600
(remote gateway appears)

ATDT bbs.example.com:23
CONNECT 9600

AT&Z0=bbs.example.com:23
OK
ATDS0
CONNECT 9600

AT&Z1=Pine.Example.com:22
OK
(note: hostname case is preserved)

ATDT 5551234
(looked up in dialup.conf, or NO CARRIER if unmapped)
```

9.2.2 Dial-up Mapping

The gateway also maintains `dialup.conf`, which maps phone numbers to `host:port` targets. The mapping table is **shared** across both ports — one file consulted by both modems — so a number you add is dialable from either Port A's or Port B's modem. Edit it through either port's Modem Emulator menu (*Configuration* → *M* → *A or B* → *D*). A built-in entry, `1001000`, always maps to the local gateway menu (equivalent to `ATDT ethernet-gateway`) and cannot be deleted. Numbers are matched by digits only — dashes, spaces, and parentheses are ignored — so `555-1234` and `5551234` hit the same entry.

9.2.3 Calling Another Port (Peer-Dial)

Peer-dial lets a modem port **call another serial port directly** and talk to the device attached to it — the gateway equivalent of dialing a friend's modem to swap files. Where `ATDT ethernet-gateway` lands on the gateway *menu*, peer-dial connects straight through to a specific port. It is **off by default**; enable `allow_peer_dial` from *Configuration* → *M Serial Configuration* → ☐ *P*, the web *Serial* config, or the GUI.

Dial another port by its **address**, `<Port>@<IP>`:

```
ATD B@192.168.1.50          (call Port B on the gateway at 192.168.1.50)
```

The address is exactly what the **Serial Gateway** menu shows (*Dial: <Port>@<ip>*), so you dial what you see. You can also just pick the target port in that menu — it has the same effect as dialing its address. What happens next depends on the **target port's mode**:

- **Modem-mode target — it rings.** The device on that port answers per its *own* AT rules: automatically after `S0` rings (`S0 = 0` disables auto-answer), or when its operator types `ATA`. This is a true dial-up call.
- **Console-mode (telnet-serial) target — it connects directly.** A console port has no modem to answer, so the bridge opens immediately (leased-line style).

Once connected it is a **transparent byte pipe** between the two devices — run a file-transfer protocol (XMODEM / YMODEM / ZMODEM / Kermit / Punter) end to end between them, exactly as you would over a real modem-to-modem call. Each port keeps its own baud rate; they need not match. (Turn PETSCII translation off — `AT+PETSCII=0` — before a binary transfer.)

Caller sees	When
CONNECT	Target answered (modem) or connected (console).
BUSY	Target modem port is already in a call.
NO ANSWER	Target rang but nobody answered within the caller's <code>S7</code> (needs <code>ATX3+</code> ; else <code>NO CARRIER</code>).
NO CARRIER	Unknown/disabled port, peer-dial off, dialing your own port, or the call failed.

LOCAL ECHO

The connection is a transparent link — neither gateway echoes the data, and (unlike dialing a host or BBS) there is no remote host echoing your keystrokes back. **Turn on local echo (half-duplex) on each terminal** to see what you type, exactly as with two terminals wired back to back. ATE does *not* help here: it only echoes AT commands in command mode, not the online data stream.

RING COUNT VS. ANSWER TIMEOUT

Auto-answer waits `S0` rings at the ~6 s ring cadence, so the gateway default `S0 = 5` takes ~30 s — longer than the caller's default `S7 = 15` wait, which would give `NO ANSWER` first (authentic real-modem behaviour). On a port you intend to be dialed, set a low `S0` (e.g. `ATS0=1`) or have the operator answer with `ATA`.

Cross-gateway (over the master/slave relay): a device on a *slave* can dial a port on its *master* — `ATD <Port>@<master-ip>`. The slave relays the call; the master resolves the address to one of its own ports and rings/connects it, bridging `device ↔ slave ↔ master ↔ port`. Requires the slave's `allow_peer_dial` on (to relay) and the master's `master_accept_relays + allow_peer_dial` on (to accept). Addressing is by IP, so master and slave need distinct addresses (normal for separate machines). Cross-gateway is symmetric: `<Port>@<slave-ip>` reaches **any** port a slave has — a **console** port connects directly, a **modem** port rings the attached device — from the master or, with the master as a crossbar, from another slave

(device ↔ slave-A ↔ master ↔ slave-B ↔ device). A slave modem port becomes dialable by announcing itself to the master (automatic when `gateway_role = slave` and `allow_peer_dial` are set).

9.3 Identification (ATI)

Command	Response
ATI / ATI0	Product ID and version: Hayes-compatible Ethernet Gateway Modem Emulator v0.6.3
ATI1	ROM checksum: 000 (virtual)
ATI2	ROM self-test: OK
ATI3	Firmware identifier: Ethernet Gateway v0.6.3 (Hayes-compatible)
ATI4	Product description: Hayes-compatible virtual modem over TCP
ATI5	OEM / country code: B00
ATI6	Link diagnostics: No link diagnostics available
ATI7	Product info: Product: ethernet-gateway (software emulator)

9.4 Reset & Profile Storage

Command	Action
ATZ	Reset to the stored profile (read from <code>egateway.conf</code>). Drops any active connection.
AT&F	Factory reset to gateway-friendly defaults. Does <i>not</i> overwrite the stored profile, but does drop any active connection (same as ATZ in that respect).
AT&W	Save current settings to <code>egateway.conf</code> . Persists echo, verbose, quiet, all 27 S-registers, X, &C, &D, &K, the +PETSCII toggle, and the four stored-number slots.
AT&V	Display current configuration — one line of mode flags, a second line listing every S-register, and a third line listing stored numbers.

9.5 Echo, Verbose, Quiet

Command	Effect
ATE0 / ATE1	Echo off / on (default on).
ATV0 / ATV1	Numeric / verbose result codes (default verbose).
ATQ0 / ATQ1	Result codes on / silent mode (default on).

In verbose mode results are text (OK , CONNECT , NO CARRIER , ERROR , RING , BUSY , NO DIALTONE , NO ANSWER). In numeric mode they are digits. Quiet mode suppresses results altogether.

9.5.1 ATX Levels

Level	Extended codes	CONNECT format
X0	Basic only; BUSY / NO DIALTONE / NO ANSWER collapse to NO CARRIER	CONNECT (code 1)
X1	Basic + baud in CONNECT	CONNECT <baud> (code per baud)
X2	Adds NO DIALTONE detection	CONNECT <baud>
X3	Adds BUSY detection	CONNECT <baud>
X4 (default)	Full extended set	CONNECT <baud>

Numeric CONNECT codes follow Hayes conventions: 1 = 300 baud, 5 = 1200, 10 = 2400, 12 = 9600, 16 = 19200, 28 = 38400, 87 = 115200. Non-standard baud rates report code 1.

9.6 DTR, Flow Control, and DCD

Command	Meaning
AT&C0	DCD always asserted.
AT&C1 (Hayes default)	DCD reflects carrier state.
AT&D0 (gateway default)	Ignore DTR entirely.
AT&D1	Drop DTR → return to command mode.
AT&D2 (Hayes default)	Drop DTR → hang up.
AT&D3	Drop DTR → hang up and reset.
AT&K0 (gateway default)	No modem-layer flow control.
AT&K1	Auto-detect (Hayes convention). Accepted and stored; like every &K mode, does not drive the wire (see warning below).
AT&K3	Bidirectional RTS/CTS hardware flow control.
AT&K4	Bidirectional XON/XOFF software flow control.
AT+PETSCII=0 (gateway default)	PETSCII translation off — direct-TCP dials (ATDT host:port) are raw ASCII passthrough.
AT+PETSCII=1	PETSCII translation on — vendor extension for C64/PET callers. The emulator case-swaps letters in both directions and maps ASCII BS ↔ PETSCII DEL so an ASCII BBS like telehack.com reads correctly. ANSI ESC [... cursor/color sequences are stripped on the inbound side so they don't render as garbage. Inbound punctuation that renders wrong on a C64 in lower/upper mode is normalized to plain ASCII: back-tick (the old “left single quote”, which shows as a thick underscore) and UTF-8 “smart” single quotes → ' ; smart double quotes → " ; tilde and en/em dashes → - ; the ellipsis → . . . ; other high bytes the C64 can't display are dropped or shown as ?. While editing the AT command line under +PETSCII=1, the C64 INST/DEL key (PETSCII DEL, 0x14) works as backspace and echoes correctly. Applies only to direct-TCP dials; ATDT ethernet-gateway already detects PETSCII terminals via the telnet menu and renders accordingly. Text only — turn it off (AT+PETSCII=0) before running a file transfer over a direct-TCP dial: it rewrites bytes in both directions and will corrupt XMODEM/YMODEM/ZMODEM/Kermit/Punter binary data. Persists immediately (and is re-saved by AT&W); also editable from the telnet, web, and GUI config surfaces.

GATEWAY-FRIENDLY DEFAULT DEVIATIONS

The emulator deviates from three Hayes defaults because retro clients rarely drive the relevant signals correctly:

- AT&D0 instead of &D2 — many C64, CP/M, and similar clients don't wire DTR. &D2 would cause spurious disconnects.
- AT&K0 instead of &K3 — retro clients rarely implement hardware flow control. Physical-wire flow is still controlled by the `serial_x_flowcontrol` setting on each port.
- S7=15 instead of S7=50 — keeps failed TCP dials responsive. Internally capped at 60 seconds.

All three can be overridden interactively (e.g. AT&D2, AT&K3, ATS7=50) and persisted with AT&W.

SIGNAL PINS ARE MOSTLY NOT DRIVEN

AT&C, AT&D, and AT&K are parsed, stored, persisted, and shown by AT&V — but the emulator drives no RS-232 hardware pins by default. RI, DSR, and the read direction of DTR are not controlled, and RTS/CTS handshaking on the wire is governed by each port's

`serial_x_flowcontrol` setting rather than &K. The one exception is carrier: enable the per-port `serial_x_drive_carrier` opt-in (default off) and the emulator drives **DTR** as a carrier proxy following AT&C — assert on CONNECT, drop on NO CARRIER under &C1 (or forced on under &C0) — which a terminal wired DTR→DCD sees as carrier detect. Most retro terminal software works fine without these signals; if your program requires DCD before communicating, enable that opt-in or look for a "force DTR" / "ignore carrier" option.

Note that `serial_x_drive_carrier` is a config-file setting, not modem state: because it reflects physical cabling it is **not** reset by ATZ/AT&F, **not** saved by AT&W, and **not** shown in AT&V (unlike AT&C). Change it in the GUI/web config or the telnet per-port modem menu () — the AT&C mode it follows is the normal, ATZ-resettable modem setting.

AT+PETSCII IS A VENDOR EXTENSION

AT+PETSCII=0/1 is not part of the original Hayes command set — it's a gateway-specific addition for PETSCII translation on direct-TCP dials. It lives in the ITU-T V.250 + extended-command namespace, the sanctioned home for vendor commands. It deliberately does *not* use AT&P: on real Hayes-compatible modems AT&Pn sets the pulse-dial make/break ratio, so reusing it would clash with that standard meaning. The command is set-only (=0/=1);

AT+PETSCII=1 persists the setting immediately, ATZ reloads the saved value, AT&F resets to off, and AT&V reports the state as +PETSCII:n.

9.7 S-Registers

Query with ATSn?, set with ATSn=v. Values are 0–255. ATs? prints a help table. AT&W saves all 27 values to `egateway.conf`; ATZ restores them from the stored profile; AT&F resets to factory defaults.

Register	Default	Description
S0	5	Auto-answer ring count (0 disables).
S1	0	Current ring counter.
S2	43	Escape character (+). Values > 127 disable escape detection.
S3	13	Carriage return character.
S4	10	Line feed character.
S5	8	Backspace character.
S6	2	Wait-for-dial-tone seconds (used by W modifier).
S7	15	Wait-for-carrier seconds. Hayes default is 50; gateway reduces it for responsive TCP dials. Capped internally at 60.
S8	2	Comma pause time (seconds).
S9	6	Carrier detect response time (1/10 s).
S10	14	Carrier loss disconnect time (1/10 s).
S11	95	DTMF tone duration (ms).
S12	50	Escape-sequence guard time (1/50 s). 50 = 1.0 s. Set to 0 to disable escape detection.
S13 – S24	0	Reserved placeholders. Stored and persisted so legacy init strings don't halt with ERROR, but they have no effect.
S25	5	DTR detect time (1/100 s). Reserved — no DTR pin.
S26	1	RTS-to-CTS delay (1/100 s). Reserved — no RTS/CTS pins.

Do NOT COLLIDE S3/S4/S5

Command-mode line editing dispatches on the raw byte and checks the CR branch before the BS branch. Setting `S3 = 8` would make backspace terminate the line. Keep S3/S4/S5 distinct (the Hayes defaults 13/10/8 do the right thing).

9.8 The +++ Escape Sequence

From online (data-transfer) mode, type `+++` surrounded by at least **S12 × 1/50** seconds of silence on either side to return to command mode without hanging up. Once in command mode, `AT0` resumes the connection and `ATH` hangs up. This follows the Hayes guard-time spec exactly.

The implementation details:

- Escape character is `S2` (default 43 = `+`).
- Guard time is `S12 / 50` seconds on both sides of the triple.
- Setting `S12 = 0` or `S2 > 127` disables escape detection entirely.
- Fewer or more than three consecutive escape characters are passed through as data.
- The escape is interrupted if the sender emits any non-escape byte mid-sequence.

ESCAPE WON'T TRIGGER? (BIT-BANGED SERIAL)

On a C64 with a software/bit-banged interface (EZ232 and similar) the wire can pick up a stray byte while “idle,” which silently defeats the escape: a phantom byte just before your first + resets the leading guard timer, and one within a second of the third + flushes +++ to the host and cancels the escape. To see exactly what happens, set EGATEWAY_GATEWAY_DEBUG=1 (or toggle Gateway Debug Trace) and try again — the log shows [esc] lines for each escape character accepted, the silence measured before the first one, and the offending byte if a sequence is broken. Lowering S12 (e.g. ATS12=25 for a half-second guard) tightens the window that a stray byte can spoil.

9.9 A/ — Repeat Last Command

A/ (capital A, forward slash) with no AT prefix and no carriage return repeats the last command you entered. Real Hayes modems recognise this as a special two-character command. Useful for re-dialling or re-sending the last ATZ while debugging.

9.10 Chained Command Lines

Multiple AT commands can be combined on a single AT line — the same way real Hayes modems have always accepted them in init strings. The emulator parses the line left-to-right, applies each subcommand, and emits exactly one OK at the end (or ERROR if any subcommand failed). Spaces between subcommands are tolerated.

```
ATE0Q1V1      ; echo off, quiet, verbose, single OK
ATE0 Q1 V1    ; same – spaces between tokens are fine
ATIE0         ; print version, then echo off
ATE0DT bbs.example.com:23 ; settings then dial (dial terminates)
AT&FE0        ; factory reset, then echo off – order matters
ATS0=2H       ; set S0=2 then hang up
```

The chain stops at the first *terminator* — anything that ends command mode or has a value that runs to end-of-line:

- D... — any dial command (ATD, ATDT, ATDP, ATDL, ATDSn) consumes the rest of the line and goes online.
- A — answer; the modem leaves command mode after this.
- 0 / 00 — return to online mode.
- &Zn=... — store-number value runs to end of line (it can contain any character — phone numbers, hostnames, ports).

The chain also stops at the first **error**. Commands before the failure run to completion (their state changes apply); commands after do not run. A line like ATS0=999E0 (S0=999 is out of range) returns ERROR and leaves the echo setting untouched.

S-register set values stop at the first non-digit, so chaining works naturally: ATS0=2E0 sets S0 to 2 and then echo off, even though there is no separator between 2 and E. Unknown

subcommands (`ATL`, `ATM`, `ATB`, etc.) do not break the chain — they are accepted silently as if they ran to no effect.

9.11 Ring Emulator

Telnet and SSH users can simulate an incoming phone call to a serial device from the per-port Modem Emulator menu (`I` — Ring emulator). The Ring item is per-port: pick *Configuration* → *M (Serial Configuration)* → *A or B* first to choose which port should ring, then press `I`. The modem sends `RING` to that port at standard US cadence (every 6 seconds). After `S0` rings (default 5) the modem auto-answers and the serial device receives the Ethernet Gateway main menu, just as if it had dialled in with `ATDT ethernet-gateway`. The serial device can also answer manually with `ATA` during ringing. The two ports' ring slots are independent — Port A and Port B can ring simultaneously.

9.12 Unknown AT Commands

Commands the emulator has no hardware to implement — `ATB`, `ATC`, `ATL` (speaker loudness), `ATM` (speaker mute), `AT&B`, `AT&G`, `AT&J`, `AT&S`, `AT&T`, `AT&Y`, etc. — are accepted and return `OK`. This keeps legacy init strings working instead of halting mid-setup with `ERROR`.

9.13 Console Mode (Telnet-Serial Bridge)

Either of the two physical serial ports can be repurposed as a transparent telnet-serial bridge — independently of the other. When a port has `serial_x_mode = console`, its Hayes emulator is disabled and that port idles until a telnet/SSH user picks **G Serial Gateway** from the main menu and chooses this port from the A/B picker. Bytes then flow in both directions between the user's session and the wire, with no AT processing. Both ports can host their own concurrent bridge — a session bridged to Port A doesn't block a separate session from bridging to Port B.

This is the right mode when a serial port is connected to something that already speaks its own protocol — a microcontroller's debug console, a router's serial console, an embedded test fixture — and you want to drive it remotely.

SWITCHING MODES (PER PORT)

- **GUI:** open the chosen port's *More...* popup from the Serial Port frame and set its *Mode* dropdown from *Modem (AT Command) Mode* to *Telnet-Serial Mode*. Saving reconfigures only the changed port; the other port keeps running.
- **Telnet/SSH menu:** from *Configuration* → *M (Serial Configuration)* → *A or B*, press **T** (*Toggle Modem/Console mode*) inside the per-port menu. The banner flips between *PORT A - MODEM EMULATOR* and *PORT A - SERIAL CONSOLE* (or B), and the *Dialup Mapping / Ring Emulator* items hide in console mode (they're modem-only features). The **T** entry is hidden from sessions connected over the modem itself, since flipping to console mode would tear down the caller's own connection before they could acknowledge — flip the mode from a telnet, SSH, or system-console session instead.
- **Config file:** set `serial_a_mode = console` (or `serial_b_mode = console`) in `egateway.conf`. The change is hot-applied within one manager-poll interval — no restart required.

9.13.1 Using the Bridge

1. From the main menu, press **G** (Serial Gateway). An A/B picker appears showing both ports' status; ineligible ports (disabled, modem mode, no device) are dimmed.
2. Press **A** or **B** to pick a port. The gateway prints that port's active framing (port, baud, data/parity/stop, flow) and asks *Connect now? (Y/N):*. Press **Y** to enter the bridge.
3. Type to the wire; the wire types back. The connection is fully 8-bit transparent and the gateway does not interpret AT commands or insert IAC sequences.
4. Press **ESC** twice in a row to leave the bridge (PETSCII **←** twice on Commodore terminals). A single **ESC** is held for one read cycle; if you type any other key after it the held **ESC** goes through to the wire, so editors that need **ESC** (vi, ed, emacs) keep working.

ONE SESSION PER PORT

The bridge is single-user **per port**. Port A and Port B each accept one bridge at a time. If a second telnet/SSH session tries to open a bridge to a port that's already in use, the second session sees *Another session is already using Port X* and is returned to the main menu. The port is released as soon as the first session leaves its bridge or disconnects. Bridges on the two ports do not interfere with each other.

SELF-BRIDGING IS BLOCKED

The Serial Gateway item is always shown; the picker is the single authority on what's eligible. A session that arrived over a serial port may bridge to a *different* port but not back to its own arrival port (that would loop your terminal to itself), so the picker marks only that one port ineligible. When nothing is available the picker says so rather than the menu dead-ending.

9.14 Master/Slave Serial Extender

A **slave** gateway can extend its serial ports to a **master** gateway over the master's SSH port, so a serial-attached device reaches the master's menu, file transfer, and dial-out as if it were attached to the master directly. **Files always land on the master.** The feature is entirely inert by default (`gateway_role = standalone`); a standalone gateway behaves exactly as it always has.

9.14.1 Configuring the Master

Enable the SSH server (`ssh_enabled = true`), set `gateway_role = master`, and turn on `master_accept_relays`. Relay connections share the existing SSH port (2222) alongside normal SSH logins — there is no per-slave configuration. Accepting relays is never implied by enabling SSH; if you set master + accept-relays but leave SSH disabled, the server logs a startup warning because relays ride the SSH server.

9.14.2 Configuring the Slave

Set `gateway_role = slave`, point `slave_master_host` / `slave_master_port` at the master (the port defaults to the SSH port, 2222), and enter `slave_master_username` / `slave_master_password` — which **must match the master's unified username/password**. The slave pins the master's SSH host key on first contact (trust-on-first-use, stored in `gateway_hosts`) and refuses a changed key. All of these settings are editable from the telnet **Configuration > S Server > M Master/Slave** sub-screen, the web config *Master/Slave* card, and the GUI Server *More...* popup.

9.14.3 How Ports Relay

- **Modem-mode port** — the slave runs the Hayes emulator locally; when the device dials, the slave bridges the call to the master, which serves its menu or dials a resolved `host:port` onward. The slave's *local* dial map resolves the number and the master performs the dial ("resolve local, dial central"). `+++` / `ATO` work normally.
- **Console-mode port** — the slave registers the port with the master; a master user reaches it from the master's **G Serial Gateway** menu, which lists local ports plus registered remote (slave) ports. A slave's own Serial Gateway menu shows its relayed console port as "→ master" (not selectable locally).

Each gateway's serial ports are configured on the gateway they are physically attached to. A slave's ports use the *slave's* own `serial_a_*` / `serial_b_*` settings (device path, baud, mode, flow control, `drive_carrier`, S-registers, `AT&W`); the master's local ports use the *master's*. The relay carries the session — the byte stream, and for a console port its label — **not** the port configuration. Set each device's baud/mode/etc. on the gateway it is wired to; there is no per-slave serial configuration on the master.

In slave mode the gateway still serves its own telnet/SSH and shows a "SLAVE mode: ports relay to master" notice with the master's address on its main menu. The slave reconnects automatically if the link drops. Only the SSH transport is implemented (`relay_transport = ssh`).

9.14.4 Relay Limitations and Operational Notes

- **A modem-mode relay reports only NO CARRIER on any failure.** Master unreachable, wrong credentials, a protocol-version skew, or a number not in the *slave's* dial map all look identical at the device — the specific cause is in the **slave's** server log, so check there when a relay dial won't connect.
- **Relay and console-registration channels count against the master's `max_sessions`.** Each active relay call *and each idle console-port registration* uses one master session slot, so size `max_sessions` to cover your slaves' registered ports plus interactive telnet/SSH/web users; a master at capacity refuses further relays.
- **Master and slave must run a compatible relay protocol version.** A skewed pair refuses to relay (the log says "upgrade the older gateway"); keep both gateways on the same release.
- **A changed master SSH host key strands a slave until you clear it.** The slave pins the master's key on first contact; if it later *changes* (e.g. the master was reinstalled) the slave refuses to reconnect as a possible MITM and cannot prompt (it is headless). Delete the master's entry from the slave's `gateway_hosts` file to let it re-pin.
- **Role and relay settings apply at server startup.** Changing `gateway_role` or `master_accept_relays` (or the slave's master address) takes effect on the next server restart, not live — the telnet Master/Slave screen shows a restart reminder; web/GUI users must restart the server themselves. An unrecognized `gateway_role` or `relay_transport` value (or `slave_master_port = 0`) is silently reset to its default.
- **A slave retries with a failure-aware backoff and won't lock out its own IP.** Network errors retry briskly (1→30 s), a wrong-credential rejection backs off ~6 min (longer than the master's 5-minute per-IP lockout window, so a misconfigured slave never trips the ban against its own host), and a relays-declined master is re-checked every ~60 s. A silently-dead link is detected within ~2 minutes via SSH keepalive.

10. SSH Gateway (Outbound)

The SSH Gateway proxies your inbound telnet or SSH session through to a remote SSH server. Useful for accessing SSH boxes from a Commodore 64 or any other client that can only speak telnet.

10.1 Using It

From the main menu, press `S`. The gateway shows the currently-configured auth mode, then prompts for the remote host, port, and username:

```
Auth: password
Host: example.com
Port (22): 22
Username: alice
```

Auth mode is set by `ssh_gateway_auth` in the config (default `password`). You can change it either from the GUI's *Server* → *More* popup or from the telnet session's *Configuration* → *Gateway Configuration* menu. The two modes are mutually exclusive — the gateway never silently falls back from one to the other, so auth failures are unambiguous:

- **Password** — the gateway asks for the remote account's password after the username prompt. Works with any SSH server that allows password auth.
- **Key** — the gateway offers its own Ed25519 keypair (`ethernet_gateway_ssh_key`, generated on first use and persisted at mode `0600`). The remote must have the matching public half in `~/.ssh/authorized_keys` first. The public key is shown in the GUI's *Server* → *More* popup when Auth is set to Key; you can also extract it on the command line with `ssh-keygen -y -f ethernet_gateway_ssh_key`.

Press `ESC` twice to disconnect.

10.2 Host-Key Trust (TOFU)

The first time you dial a given host, the gateway shows the server's SHA-256 fingerprint and asks whether to trust it. Accepted fingerprints are persisted to `gateway_hosts` (mode `0600`) and checked on every subsequent dial. A changed key triggers a prominent warning:

```
WARNING: HOST KEY HAS CHANGED!
This could indicate a security threat.
New type: ssh-ed25519
New fingerprint: SHA256:...
Update key? (Y/N):
```

All trust decisions — first accept, update, and reject — are written to the server log (and the GUI console) so you have a forensic trail if a changed key turns out to be a man-in-the-middle attempt.

10.3 Output Filtering

For PETSCII and ASCII terminals, ANSI escape sequences from the remote are automatically stripped and text is converted to the appropriate encoding. ANSI terminals get the raw output unmodified. PTY size is set to 40×25 for PETSCII and 80×24 for ANSI/ASCII.

10.4 Key File Summary

File	Role	Analogous to
gateway_hosts	Remote host fingerprints accepted by the SSH Gateway.	~/.ssh/known_hosts
ethernet_ssh_host_key	Host key the SSH <i>server</i> presents to incoming clients.	/etc/ssh/ ssh_host_ed25519_key
ethernet_gateway_ssh_key	Outbound SSH client keypair used for pubkey auth to remote hosts.	~/.ssh/id_ed25519

11. Telnet Gateway (Outbound)

Press **T** from the main menu to dial a remote telnet server or any raw-TCP service on port 23 (BBSes, MUDs, legacy hand-rolled software).

11.1 Protocol Mode

The gateway displays the currently-configured mode on entry and proceeds straight to the host prompt. The mode is a persistent setting, not a per-session choice:

```
Mode: Telnet protocol
Host:
```

Change the mode from the GUI (*Server* → *More* popup, Telnet Gateway Mode dropdown) or from the telnet session's *Configuration* → *Gateway Configuration* menu. Changes take effect on the next dial — no restart needed.

11.2 The Three Modes Explained

11.2.1 Reactive mode (default)

`telnet_gateway_negotiate = false` and `telnet_gateway_raw = false`. The gateway parses IAC in both directions but does not send proactive option offers. It silently accepts cooperative `WILL ECHO` from the remote (so BBSes that expect the server to echo work correctly), refuses every other peer-initiated option with `DONT / WONT`, escapes outbound `0xFF` data bytes as `IAC IAC`, and parses inbound IAC so protocol bytes don't leak to the user's terminal. Works with any real telnet server *and* with raw-TCP services on port 23 that never initiate telnet options.

11.2.2 Cooperative mode

`telnet_gateway_negotiate = true`. In addition to reactive mode's behaviour, the gateway sends proactive `IAC WILL TTYPE`, `IAC WILL NAWS`, and `IAC DO ECHO` at connect time. On `SB TTYPE SEND` it replies with the user's terminal type (`PETSCII`, `ANSI`, or `DUMB`). On `DO NAWS` it replies with the user's actual window dimensions, and it forwards in-session `NAWS` updates if the user resizes their terminal. Option state is tracked through a full RFC 1143 six-state Q-method, so a mind-change while a prior offer is in flight resolves cleanly.

Enable this when dialling modern BBSes — you get echo cooperation, correct terminal-type adaptation, and full-screen UI at your actual window size.

11.2.3 Raw TCP

`telnet_gateway_raw = true`. The gateway disables its entire telnet-IAC layer: no IAC escaping on outbound, no IAC parsing on inbound. Supersedes `telnet_gateway_negotiate`. Intended for destinations that aren't really telnet — some legacy MUDs, hand-rolled BBS software, or services that happen to run on port 23 but speak some other byte-level protocol.

Bytes written *to* the local user are still IAC-escaped so their telnet client does not misinterpret a stray 0xFF as a protocol byte.

11.3 Dialing

After the mode prompt, enter the host and port:

```
Host: bbs.example.com
Port (23): 23
```

Press `ESC` twice to disconnect. For PETSCII and ASCII terminals, ANSI escape sequences from the remote are filtered out. The gateway also has an 8 KiB cap on subnegotiation body length, so a malicious remote cannot exhaust memory by sending huge `SB` bodies without a terminating `IAC SE`.

12. AI Chat

The AI Chat feature sends prompts to the Groq API and displays the responses in the terminal. It is free to use — you need a Groq account and an API key — and the model is `llama-3.3-70b-versatile`.

12.1 Setup

1. Go to `https://console.groq.com/keys` and create a free account.
2. Generate an API key. It starts with `gsk_`.
3. In the GUI, paste the key into the *API Key* field (it displays as dots), or edit `egateway.conf` and set `groq_api_key = gsk_your_key_here`.
4. Save. No restart needed.

12.2 Using AI Chat

From the main menu, press `A`. Type a question and press Enter. The server shows *Thinking...* while waiting for the response (timeout 30 seconds). The answer is paginated:

- `N` / `P` — next / previous page.
- `Q` — return to the main menu.
- From the answer screen, type a new question to continue the conversation.

Responses are word-wrapped to fit the terminal — 38 columns for PETSCII, 78 for ANSI/ASCII. Long code blocks and unusual punctuation are preserved but may wrap awkwardly on narrow screens.

No API KEY?

If `groq_api_key` is empty and the user presses `A`, the gateway displays instructions for obtaining a key instead of attempting the API call.

13. Web Browser

The text-mode browser renders HTML as plain text with numbered link markers. It works on all terminal types, including 40-column PETSCII.

13.1 Opening a Page

From the main menu, press **B**. The homepage loads automatically if `browser_homepage` is set (default `http://telnetbible.com`). You can also enter a URL or search query:

- A URL without a scheme — `example.com` — gets `https://` prepended automatically.
- Text without dots is treated as a search query and sent to DuckDuckGo.
- A `gopher://` URL uses the Gopher protocol.

13.2 Following Links

Links are marked with numbered tags: `[1]`, `[2]`, `[3]`. Press **L** and enter a number to follow.

13.3 Navigation Commands

Key	Action
N / P	Next / previous page.
T / E	Jump to top / end.
L	Follow link by number.
G	Go to a new URL or search.
S	Search for text in the page.
F	Fill out and submit forms.
K	Save as bookmark; on the home screen, opens the bookmarks list.
B	Back to the previous page.
R	Reload.
H	Help.
Q	Close page (back to browser home).
ESC	Exit to main menu.

13.4 Bookmarks and Forms

Up to 100 bookmarks are saved in `bookmarks.txt` next to the binary. When a page has forms, press **F** to interact — pick a form, edit fields by number, **S** to submit or **Q** to cancel.

13.5 Limits

- Maximum page size: 1 MB.
- Maximum rendered lines: 5000.
- HTTP request timeout: 15 s.
- Page history depth: 50.
- HTTPS connections that fail due to TLS errors retry over HTTP.

14. Weather

The Weather feature displays the current conditions and a 3-day forecast for any US ZIP code. Data is pulled from Open-Meteo (<https://open-meteo.com>), which is free and requires no API key.

1. From the main menu, press **W**.
2. Enter a 5-digit US ZIP code, or press Enter to accept the last-used one.
3. Current temperature, humidity, wind speed and compass direction, plus a 3-day forecast are displayed.

The last-used ZIP is persisted to `weather_zip` in `egateway.conf` so subsequent calls default to the same location.

15. Signal Handling & Operations

The gateway installs POSIX signal handlers at startup using the `signal-hook` crate. On receipt of any of `SIGINT`, `SIGTERM`, or `SIGHUP`, the server begins an orderly shutdown.

Signal	Typical source	Effect
SIGINT	<code>Ctrl+C</code> from the terminal	Shut down. Connected sessions are notified with a broadcast message.
SIGTERM	<code>kill</code> , <code>systemd</code>	Shut down.
SIGHUP	<code>systemd reload</code> , terminal disconnect	Shut down. (The unit file maps <i>reload</i> to SIGHUP for familiar semantics.)

15.1 Shutdown Sequence

When a signal arrives:

1. An `AtomicBool` flag is set.
2. A watcher thread fires a `tokio Notify` to wake the async runtime.
3. Listeners stop accepting new connections.
4. A broadcast notification is sent to every active telnet, SSH, and master/slave relay session:
"Server shutting down. Goodbye."
5. The serial modem thread emits the same notice and exits cleanly, releasing the port.
6. The GUI window closes automatically.
7. The process exits.

15.2 Under systemd

```
sudo systemctl status ethernet-gateway    # current status
journalctl -u ethernet-gateway -f         # live log
sudo systemctl reload ethernet-gateway    # graceful stop (SIGHUP)
sudo systemctl restart ethernet-gateway   # restart
sudo systemctl stop ethernet-gateway      # stop (SIGTERM)
```

The unit's `Restart=on-failure` policy respawns the process if it exits with a non-zero code, rate-limited to 5 starts per 60 seconds via `StartLimitIntervalSec` / `StartLimitBurst`.

16. Troubleshooting

16.1 Telnet Session Refused (Public IP)

If an unauthenticated client sees "Connections from this address are not accepted", the source IP is outside the private ranges listed in chapter 5. Either connect from a private-network address or enable `security_enabled = true` and configure credentials.

16.2 Session Denied — Too Many Attempts

Per-IP lockout after three failures. Wait 5 minutes. The lockout covers both telnet and SSH — retrying on the other protocol will not reset the counter.

16.3 No Serial Port Detected

- Plug in the adapter before clicking the refresh (↻) button in the GUI.
- On Linux, make sure your user is in the `dialout` group (`sudo usermod -aG dialout "$USER"`), then log out and back in.
- On Linux under systemd, the unit file sets `PrivateDevices=yes` which hides most device nodes. If you need USB-serial access, remove that hardening line or switch to a raw `DeviceAllow=` rule for your exact adapter.
- On macOS, adapters typically show up as `/dev/tty.usbserial-XXXX` or `/dev/tty.usbmodemXXXX`.
- On Windows, check Device Manager. Ports appear as `COM3`, `COM4`, etc.

16.4 File Transfer Times Out

Turn on verbose mode (`verbose = true` in `egateway.conf`, or the *Verbose Transfer Logging* checkbox in the GUI General frame) and retry. The log will show every header byte, every CRC result, every timeout, tagged with the protocol (`XMODEM recv:`, `ZMODEM send:`, etc.). Common causes:

- Client and server disagree on CRC vs. checksum — the client is very old and never implemented CRC. The gateway falls back to checksum automatically after two-thirds of the negotiation timeout.
- IAC escaping mismatch when transferring binary files with `0xFF` bytes over telnet. Toggle it with ☐ in the File Transfer menu.
- Slow or unreliable serial link — raise `xmodem_block_timeout` and `xmodem_max_retries` for XMODEM/YMODEM, `zmodem_frame_timeout` and `zmodem_max_retries` for ZMODEM, or `punter_block_timeout` and `punter_max_retries` for Punter (and lower `punter_block_size` toward ~40 on a noisy line).
- Client is slow to start after you pick the protocol — raise `xmodem_negotiation_timeout` (XMODEM/YMODEM), `zmodem_negotiation_timeout` (ZMODEM), or `punter_negotiation_timeout` (Punter).

- Log shows duplicate block 0 retransmissions during YMODEM setup — raise `xmodem_negotiation_retry_interval` toward the spec-recommended 10 s so fewer `C` pokes stack up in the sender's buffer before it starts responding.

16.5 SSH Won't Connect

- Check `ssh_enabled = true` and that the server has been restarted since the change.
- Confirm the port with `ss -tlnp | grep 2222` on Linux.
- On first connect, accept the gateway's host-key fingerprint. Your client records it in `~/.ssh/known_hosts`.
- If your client complains about host-key mismatch, you have probably regenerated `ethernet_ssh_host_key`. Delete the old entry from `known_hosts` or compare fingerprints carefully.
- Public-key auth against the *server* is not supported — only password auth. Public-key auth is only available on the *outbound* SSH Gateway.

16.6 Modem Shows "NO CARRIER" on Every Dial

- Check the host and port — `ATDT host:port` must include the port unless the host resolves to something listening on 23.
- S7 defaults to 15 seconds. Networks that need longer dial times should `ATS7=50` (or up to 60).
- Verify the host is reachable from the gateway machine with `nc -zv host port` or `telnet host port`.
- If you dialled a phone number (all-digit string), check `dialup.conf` — unmapped numbers return `NO CARRIER`.

16.7 GUI Won't Open

- Check that `enable_console = true`.
- On headless servers, leave it at `false` — the GUI requires a display server (X11 or Wayland).
- On Linux, make sure `libudev`, `libxkbcommon`, and your display-server libraries are installed.
- Missing GPU support? The GUI uses `wgpu` via `eframe`. Software-only fallback via a different feature set would require a rebuild; most modern Linux distros work out of the box.

16.8 Stuck Serial Session After Parameter Change

If you change baud from a serial session with mismatched terminal settings, don't panic. The gateway waits up to 60 seconds for confirmation and reverts automatically on timeout. Within 60 seconds your old settings come back.

16.9 Garbled Input from a Commodore 64 (Bit-Banged Serial)

Symptom: a command typed on a C64 over a userport RS-232 adapter runs *truncated or corrupted* — usually a single typed character arrives as a stray PETSCII graphics byte — and it tends to happen on *longer* commands that wrap past the 40-column screen edge. It is intermittent: the same command may go through cleanly on the next try.

Cause: passive userport adapters like the **EZ232** have no UART; the C64 times every serial bit in software (“bit-banging”), which has very little timing margin. When a long line wraps, the screen scrolls and the KERNAL briefly masks interrupts to move screen memory — long enough to disrupt the bit timing of a keystroke being transmitted at that moment, garbling the byte on the wire. This is a C64 transmit-side timing fault that happens *before* any byte reaches the gateway, so it affects the SSH Gateway, the Telnet Gateway, and every other feature that rides the same serial link.

Fix:

- **Use 1200 baud, not 2400**, on bit-banged userport adapters. Set `serial_x_baud = 1200` on the gateway port *and* set your C64 terminal program (e.g. CCGMS) to 1200. This roughly doubles the per-bit timing budget and is the reliable cure.
- If you must run faster, a hardware-UART interface (SwiftLink / Turbo232) or a WiFi modem with its own UART removes the bit-banging entirely and is not affected.
- To confirm the diagnosis on your own hardware, enable the gateway byte-trace (§16.10) and watch the per-byte timing and the assembled `[gw-in]` line.

16.10 Gateway Byte-Trace (Debug Mode)

The SSH Gateway and Telnet Gateway can log every byte that crosses the gateway, which is useful for diagnosing input corruption, terminal translation, and flow problems like §16.9. It costs nothing when off and needs no special build.

Enable it with the `gateway_debug` setting, which you can toggle from any of three places without restarting the program:

- **GUI:** the *General* frame — tick *Gateway Debug Trace* (next to *Show GUI on Startup*) and click *Save*.
- **Web console:** the *General* frame — same checkbox, then *Save*.
- **In-session telnet menu:** *Configuration* → *O Other Settings* → *D*, or *Configuration* → *M Serial Configuration* → *D*. Both toggle the same flag.

The flag is read fresh when each gateway session starts, so a change takes effect on the *next* SSH/Telnet Gateway connection — no restart needed. It is persisted to `egateway.conf` as `gateway_debug = true`.

FORCING IT ON FROM THE ENVIRONMENT

The `EGATEWAY_GATEWAY_DEBUG` environment variable still forces the trace on regardless of the config flag — handy for headless or scripted runs where you want it active from the moment the process starts:

```
EGATEWAY_GATEWAY_DEBUG=1 ./ethernet-gateway
```

Under `systemd`, add `Environment=EGATEWAY_GATEWAY_DEBUG=1` to the unit. When set, it overrides the `gateway_debug` setting.

Where the trace appears: the same places as all gateway log output — the process's `stderr`, the GUI's live console, and (when `web_enabled = true`) the web-config console's `/logs` endpoint, e.g. `curl http://host:8080/logs`.

Reading the trace. When a gateway session opens you'll see a banner, then one line per input byte and a summary line per typed command:

Line	Meaning
[gw] SSH gateway trace ON - ...	Session opened (or Telnet gateway); shows terminal type and PTY/window size.
[gw-in] +Δms t=...ms byte=0xNN 'c' (petSCII 0xNN)	One byte forwarded to the remote. +Δms is the gap since the previous byte; t= is elapsed since the trace started; the petSCII note appears only when the PETSCII→ASCII case-swap changed the byte.
[gw-in] line (N bytes) -> hex "ascii"	The full line as the remote shell received it, flushed on RETURN (CR/LF). This is exactly what bash ran.
[gw-out] raw N bytes -> ... / filtered ...	Bytes received from the remote, shown before and after the PETSCII/ASCII output filter.

The `+Δms` timing distinguishes terminal behaviours: large per-byte gaps mean the client sends each keystroke live (character mode, e.g. CCGMS), while a near-zero burst of bytes means a whole line was sent at once. This is verbose, per-byte output intended for troubleshooting — leave it disabled in normal operation.

Appendix A — Configuration Key Reference

Every key recognised by `egateway.conf`, sorted alphabetically, with two intentional deferrals: the `kermit_*` protocol-tuning family (everything *except* `kermit_server_enabled` and `kermit_server_port`, which are listed here because they affect the gateway's listening surface) is documented in the standalone `kermit.html` reference, and §3.2 of this manual carries one-line summaries of the entire family for at-a-glance scanning. The parser is case-sensitive on keys but booleans (`true/false`) are compared case-insensitively. Unknown keys are silently ignored. Out-of-range numeric values fall back to the default.

Key	Type	Default	Range / validation	Description
allow_atdt_kermit	bool	false	true/false	Permit ATDT KERMIT (and the al server / ATDT kermit server) straight into Kermit server mod by default because it bypasses
browser_homepage	string	http://telnetbible.com	any URL or empty	Page loaded when the browser prompt.
disable_ip_safety	bool	false	true/false	Remove the private-IP allowlist when security_enabled is fal without auth. Use only on isolat exposed deployment, enable se
enable_console	bool	true	true/false	Show GUI window at startup. S
gateway_debug	bool	false	true/false	Byte-level trace of SSH/Telnet corruption / terminal-translatio toggling takes effect on the nex EGATEWAY_GATEWAY_DEBUG enviro regardless. See §16.10.
groq_api_key	string	empty	any	Groq API key. Empty disables A file.
idle_timeout_secs	u64	900	0 = disabled	Per-session idle timeout. Applie
kermit_server_enabled	bool	false	true/false	Bind a dedicated TCP listener t straight into Kermit server mod gate (same risk profile as allow standalone kermit.html refere documented in kermit.html.
kermit_server_port	u16	2424	≥ 1	TCP port for the standalone Ker
max_sessions	usize	50	≥ 1	Concurrent telnet-session cap.
password	string	changeme	non-empty	Unified login password — share web UI. Stored plaintext.
punter_block_size	u16	255	8-255	Punter C1 total block size in by payload at the default). Lower t retransmit unit.
punter_block_timeout	u64	20	≥ 1 (seconds)	Per-block read timeout once a R
punter_max_retries	u32	10	≥ 1	Retry cap per block (BAD / SYN
punter_max_bad_rounds	u32	30	≥ 1	Consecutive corrupt-block rese (kept higher than punter_max_ resends, so a low value strands
punter_hangup_on_failure	bool	false	true / false	

Key	Type	Default	Range / validation	Description
				Drop carrier on give-up so a str sees loss-of-carrier instead of h default.
punter_negotiation_retry_interval	u64	5	≥ 1 (seconds)	Gap between handshake-code r
punter_negotiation_timeout	u64	45	≥ 1 (seconds)	Seconds to wait for the Punter
security_enabled	bool	false	true/false	Require login on telnet and SSH telnet.
serial_x_baud	u32	9600	≥ 300	Serial baud rate. x = a or b (per
serial_x_databits	u8	8	5, 6, 7, or 8	Data bits.
serial_x_dcd_mode	u8	1	0 or 1	Saved AT&C DCD mode.
serial_x_dtr_mode	u8	0	0-3	Saved AT&D DTR-handling mode
serial_x_echo	bool	true	true/false	Saved ATE echo state.
serial_x_enabled	bool	false	true/false	Activate this port. Pairs with se independently.
serial_x_flow_mode	u8	0	0-4	Saved AT&K modem-layer flow r
serial_x_mode	string	modem	modem or console	modem = Hayes AT emulator on via G Serial Gateway as a tel
serial_x_flowcontrol	string	none	none / hardware / software	Wire-level flow control for this p
serial_x_parity	string	none	none / odd / even	Parity bit.
serial_x_petscii_translate	bool	false	true/false	Saved AT+PETSCII state — PET (Appendix B). Editable from the GUI config surfaces.
serial_x_port	string	empty	OS-specific path	Serial device. /dev/ttyUSB0, C
serial_x_quiet	bool	false	true/false	Saved ATQ quiet mode.
serial_x_s_regs	string	see note	CSV of 27 u8 values	S0-S26, comma-separated. Def 5,0,43,13,10,8,2,15,2,6,14,
serial_x_stopbits	u8	1	1 or 2	Stop bits.
serial_x_stored_0 ... serial_x_stored_3	string	empty	any	Stored phone-number slots, per

Key	Type	Default	Range / validation	Description
serial_x_verbose	bool	true	true/false	Saved ATV verbose/numeric mode.
serial_x_x_code	u8	4	0–4	Saved ATX result-code level.
ssh_enabled	bool	false	true/false	Run the SSH server.
ssh_gateway_auth	string	password	key or password	Auth mode for the outbound SSH server.
ssh_port	u16	2222	≥ 1	SSH listen port.
telnet_enabled	bool	true	true/false	Run the telnet server.
telnet_gateway_negotiate	bool	false	true/false	Outbound telnet gateway: send negotiation offers.
telnet_gateway_raw	bool	false	true/false	Outbound telnet gateway: raw-mode.
telnet_port	u16	2323	≥ 1	Telnet listen port.
transfer_dir	string	transfer	non-empty	Base directory for file transfers.
username	string	admin	non-empty	Unified login username — shared with the web UI.
verbose	bool	false	true/false	Detailed per-block / per-subpacket logging: YMODEM, ZMODEM, Kermit, and XMODEM.
weather_zip	string	empty	5 digits	Last-used US ZIP. Updated automatically.
web_enabled	bool	false	true/false	Run the configuration web server. Disables the GUI settings page in a browser.
web_port	u16	8080	≥ 1	Web-server listen port.
xmodem_block_timeout	u64	20	≥ 1 (seconds)	Per-block timeout during transfer.
xmodem_max_retries	usize	10	≥ 1	NAK retry cap per block.
xmodem_negotiation_retry_interval	u64	7	≥ 1 (seconds)	Gap between C/NAK pokes during negotiation.
xmodem_negotiation_timeout	u64	45	≥ 1 (seconds)	Initial negotiation timeout for XMODEM.
zmodem_frame_timeout	u64	30	≥ 1 (seconds)	Per-frame read timeout during transfer.
zmodem_max_retries	u32	10	≥ 1	Retry cap for ZRQINIT / ZRPOS.
zmodem_negotiation_retry_interval	u64	5	≥ 1 (seconds)	Gap between ZRINIT / ZRQINIT pokes during handshake.
zmodem_negotiation_timeout	u64	45	≥ 1 (seconds)	Initial ZRQINIT / ZRINIT handshake timeout.

Appendix B — AT Command Reference

Quick reference for every AT command the emulator understands. See chapter 9 for detailed explanations. All commands are case-insensitive. Multiple commands may be combined on a single AT line — see [section 9.10](#) for the chaining rules (ATE0Q1V0, ATE0DT host, ...).

B.1 Core Commands

Command	Effect
AT	Attention. Returns OK.
AT?	Display AT command help.
ATZ	Reset to stored profile (saved by AT&W).
AT&F	Factory reset to gateway-friendly defaults.
AT&W / AT&W0	Save current settings to egateway.conf.
AT&V	Display current configuration.
ATH / ATH0	Hang up any active connection.
ATA	Answer incoming ring.
ATO / AT00	Return to online mode (after +++ escape).
A/	Repeat last command (no AT prefix, no CR).
+++	Escape to command mode (guard-time gated).

B.2 Identification

Command	Response
ATI / ATI0	Product ID and version.
ATI1	ROM checksum (000).
ATI2	ROM self-test (OK).
ATI3	Firmware identifier.
ATI4	Product description.
ATI5	OEM / country code (B00).
ATI6	Link diagnostics.
ATI7	Product info.

B.3 Echo, Verbosity, Quiet, Result Codes

Command	Effect
ATE0	Echo off.
ATE1 (default)	Echo on.
ATV0	Numeric result codes.
ATV1 (default)	Verbose (text) result codes.
ATQ0 (default)	Result codes enabled.
ATQ1	Quiet mode — all results suppressed.
ATX0	Basic result codes only.
ATX1	Basic plus baud rate in CONNECT.
ATX2	Adds NO DIALTONE detection.
ATX3	Adds BUSY detection.
ATX4 (default)	Full extended result codes.

B.4 Hardware Pin State

Command	Effect
AT&C0	DCD always asserted.
AT&C1 (Hayes default)	DCD reflects carrier state.
AT&D0 (<i>gateway default</i>)	Ignore DTR.
AT&D1	DTR drop → command mode.
AT&D2 (Hayes default)	DTR drop → hang up.
AT&D3	DTR drop → hang up + reset.
AT&K0 (<i>gateway default</i>)	No modem-layer flow control.
AT&K1	Auto-detect (stored only; no wire-level effect).
AT&K3	Bidirectional RTS/CTS.
AT&K4	Bidirectional XON/XOFF.

B.5 Dialing

Command	Effect
ATD <i>target</i>	Dial <i>target</i> . Bare D (no T/P).
ATDT <i>host:port</i>	Tone-dial. Default for all TCP dials.
ATDP <i>host:port</i>	Pulse-dial (identical to ATDT over TCP).
ATDT ethernet-gateway	Built-in shortcut: connect to this gateway's main menu.
ATDT KERMIT	Drop into Kermit server mode (requires <code>allow_atdt_kermit = true</code>). Aliases: <code>kermit</code> , <code>kermit-server</code> , <code>kermit server</code> .
ATDL	Redial last number.
ATDS / ATDS0	Dial stored number in slot 0.
ATDS1 ... ATDS3	Dial stored number in slots 1-3.
AT&Zn=s	Store phone number or host in slot <i>n</i> (0-3). Preserves hostname case.

Dial-string modifiers (phone-number context only, except `;` which applies to hostnames too):

Modifier	Effect
,	Pause for S8 seconds per comma.
W	Wait for dial tone (adds S6 seconds).
;	Return to command mode after connect.
*, #	DTMF digits, preserved for dial-up lookup.
P, T, @, !	Ignored.

B.6 S-Registers

Command	Effect
ATS?	Show S-register help.
ATS <i>n</i> ?	Query S-register <i>n</i> (returns 3-digit decimal).
ATS <i>n</i> = <i>v</i>	Set S-register <i>n</i> to <i>v</i> (0-255).

Full register list is in chapter 9 (section 9.7).

B.7 Unknown Commands

Unrecognised AT commands return `OK` (silent-accept fallback). This covers commands the emulator has no hardware to implement: `ATB`, `ATC`, `ATL` (speaker loudness), `ATM` (speaker mute), `AT&B`, `AT&G`, `AT&J`, `AT&S`, `AT&T`, `AT&Y`, and others. The intent is to let legacy init strings run to completion rather than halt mid-setup with `ERROR`.

